

$\tau_0: C_0 \rightarrow C'_0$ is a map such that $\partial'_0 \tau_0 \partial_1 = 0$, then τ_0 extends to a chain map $\tau: C \rightarrow C'$.

PROOF. Since C is projective and C' is acyclic, we can construct $(\tau_i)_{i \geq 0}$ as in the proof of the fundamental lemma I.7.4 so as to commute with boundary operators; indeed, the condition $\partial'_0 \tau_0 \partial_1 = 0$ is exactly what is needed to start the inductive construction, cf. I.7.3a. Similarly, since C is acyclic and admissible (by 3.2) and C' is relatively injective, $(\tau_i)_{i \geq 0}$ can be extended to negative dimensions by 2.4. $\square$

(5.11) Lemma. *Let C be an admissible acyclic complex of $\mathbb{Z}G$ -modules. For any projective $\mathbb{Z}G$ -module P , the complex $P \otimes C$ (with diagonal G -action) is contractible as a complex of $\mathbb{Z}G$ -modules.*

PROOF. It suffices to prove this for $P = \mathbb{Z}G$. By III.5.7, we have $\mathbb{Z}G \otimes C$ isomorphic to the induced complex $\mathbb{Z}G \otimes C'$, where C' is C regarded as a complex of abelian groups. Since C' is contractible by hypothesis, it follows that $\mathbb{Z}G \otimes C$ is contractible. $\square$

In view of 5.9 and 5.10, the construction of $\Delta: F \rightarrow F \hat{\otimes} F$ is reduced to the construction of a map $\alpha = (\alpha_p): F_0 \rightarrow \prod_{p \in \mathbb{Z}} F_p \otimes F_{-p}$ such that (i) $\partial \alpha|_{B_0} = 0$ and (ii) $(\varepsilon \otimes \varepsilon)\alpha_0 = \varepsilon$, where $B_0 \subset F_0$ is the module of boundaries. Let $\partial'_{pq} = d_p \otimes F_q: F_p \otimes F_q \rightarrow F_{p-1} \otimes F_q$ and let $\partial''_{pq} = (-1)^p F_p \otimes d_q: F_p \otimes F_q \rightarrow F_p \otimes F_{q-1}$. Then $\partial \alpha: F_0 \rightarrow \prod_{p \in \mathbb{Z}} F_{p-1} \otimes F_{-p}$ has components $\partial'_p \alpha_p + \partial''_{p-1, 1-p} \alpha_{p-1}$. So (i) is equivalent to (i') $(\partial' \alpha_p + \partial'' \alpha_{p-1})|_{B_0} = 0$, where we have omitted the subscripts on ∂' and ∂'' to simplify the notation. We now start constructing α by taking $\alpha_0: F_0 \rightarrow F_0 \otimes F_0$ to be any map satisfying (ii); this is possible because F_0 is projective. Assuming that $p > 0$ and that α_{p-1} has been defined, we wish to define $\alpha_p: F_0 \rightarrow F_p \otimes F_{-p}$ so that the diagram

$$\begin{array}{ccccc}
 & & B_0 & & \\
 & \swarrow \alpha_p|_{B_0} & \downarrow \beta|_{B_0} & & \\
 F_p \otimes F_{-p} & \xrightarrow{\partial'} & F_{p-1} \otimes F_{-p} & \xrightarrow{\partial'} & F_{p-2} \otimes F_{-p}
 \end{array}$$

commutes, where $\beta = -\partial'' \alpha_{p-1}$. I claim that $\partial' \beta|_{B_0} = 0$. In fact, if $p > 1$ then we can assume inductively that $(\partial' \alpha_{p-1} + \partial'' \alpha_{p-2})|_{B_0} = 0$, so that on B_0 we have

$$\begin{aligned}
 \partial' \beta &= -\partial' \partial'' \alpha_{p-1} && \text{by definition of } \beta \\
 &= \partial'' \partial' \alpha_{p-1} && \text{because } \partial' \text{ and } \partial'' \text{ anti-commute} \\
 &= -\partial'' \partial'' \alpha_{p-2} && \text{by the inductive hypothesis} \\
 &= 0;
 \end{aligned}$$

and if $p = 1$ then on B_0 we have

$$\begin{aligned}
 \partial' \beta &= -\partial' \partial'' \alpha_0 && \text{by definition of } \beta \\
 &= -(d_0 \otimes d_0) \alpha_0 && \text{by definition of } \partial' \text{ and } \partial'' \\
 &= -(\eta \otimes \eta)(\varepsilon \otimes \varepsilon) \alpha_0 && \text{because } d_0 = \eta \varepsilon \\
 &= -(\eta \otimes \eta) \varepsilon && \text{by (ii)} \\
 &= 0 && \text{because } \varepsilon|_{B_0} = 0.
 \end{aligned}$$

This proves the claim. By 5.11 the complex $(F_* \otimes F_{-p}, \partial')$ is contractible. We can therefore choose a contracting homotopy h and set $\alpha_p = h\beta$ to complete the inductive step. A similar argument constructs α_p for $p < 0$ by descending induction, and the proof of 5.6 is complete. $\square$

Finally, we remark that there are also *cap products* in the Tate theory. No new difficulties arise here, so we confine ourselves to a brief discussion. There is a map

$$\mathcal{H}om_G(F, M) \rightarrow \mathcal{H}om((F \hat{\otimes} F) \otimes_G N, F \otimes_G (M \otimes N))$$

given by $u \mapsto \text{id} \hat{\otimes} u \otimes \text{id}_N$. This corresponds to a map

$$\gamma: \mathcal{H}om_G(F, M) \otimes ((F \hat{\otimes} F) \otimes_G N) \rightarrow F \otimes_G (M \otimes N).$$

We now define the cap product

$$\mathcal{H}om_G(F, M) \otimes (F \otimes_G N) \rightrightarrows F \otimes_G (M \otimes N)$$

to be γ composed with

$$\text{id} \otimes (\Delta \otimes \text{id}): \mathcal{H}om_G(F, M) \otimes (F \otimes_G N) \rightarrow \mathcal{H}om_G(F, M) \otimes ((F \hat{\otimes} F) \otimes_G N),$$

where $\Delta: F \rightarrow F \hat{\otimes} F$ is any complete diagonal approximation. This induces a well-defined cap product

$$\hat{H}^* \otimes \hat{H}_* \rightarrow \hat{H}_*$$

with the usual properties.

EXERCISES

1. Show that the cup product on $\hat{H}^*$ is compatible with that defined in Chapter V on H^* . More precisely, we have a natural map $H^* \rightarrow \hat{H}^*$ which is an isomorphism in positive dimensions and an epimorphism in dimension 0; show that this map preserves products.
2. Let $\hat{H}^*(G)_{(p)}$ be the p -primary component of $\hat{H}^*(G) = \hat{H}^*(G, \mathbb{Z})$, so that we have $\hat{H}^*(G) = \bigoplus_{p||G|} \hat{H}^*(G)_{(p)}$ by the analogue of III.10.2.
 - (a) Show that each $\hat{H}^*(G)_{(p)}$ is an ideal in $\hat{H}^*(G)$, hence so is $\bigoplus_{q \neq p} \hat{H}^*(G)_{(q)}$. Consequently, $\hat{H}^*(G)_{(p)}$ is a quotient ring of $\hat{H}^*(G)$ via the projection $\hat{H}^*(G) \rightarrow \hat{H}^*(G)_{(p)}$. [Note: $\hat{H}^*(G)_{(p)}$ is not a subring of $\hat{H}^*(G)$; indeed, although the inclusion $\hat{H}^*(G)_{(p)} \hookrightarrow \hat{H}^*(G)$ does preserve products, it does not preserve identity elements.]

(b) Show that there is a ring isomorphism $\hat{H}^*(G) \cong \prod_{p||G|} \hat{H}^*(G)_{(p)}$, where each factor on the right is a ring via (a), and multiplication in the product is done componentwise.

6 Composition Products

As in ordinary cohomology (§V.4) there is a second method of defining products, based on composition of chain maps. We begin with the analogue of V.4.1:

(6.1) Proposition. *Let G be a finite group, let F be an acyclic chain complex of projective $\mathbb{Z}G$ -modules, and let $\varepsilon': F' \rightarrow \mathbb{Z}$ be a complete resolution.*

(a) *For any G -module M , the map $\varepsilon' \otimes M: F' \otimes M \rightarrow M$ induces a weak equivalence $\mathcal{H}om_G(F, F' \otimes M) \rightarrow \mathcal{H}om_G(F, M)$. In particular, if F is a complete resolution then $\hat{H}^*(G, M) \approx H^*(\mathcal{H}om_G(F, F' \otimes M)) = [F, F' \otimes M]^*$.*

(b) *If F is of finite type then $\varepsilon' \otimes M$ induces a weak equivalence*

$$F \hat{\otimes}_G (F' \otimes M) \rightarrow F \hat{\otimes}_G M = F \otimes_G M.$$

In particular, if F is a complete resolution then

$$\hat{H}_*(G, M) \approx H_*(F \hat{\otimes}_G (F' \otimes M)).$$

PROOF. For (a) we must prove $[F, F' \otimes M]_n \cong [F, M]_n$ for all $n \in \mathbb{Z}$. Replacing F by its n -fold suspension $\Sigma^n F$, we reduce to the case $n = 0$; thus it suffices to show $[F, F' \otimes M] \cong [F, M]$. Let $u: F \rightarrow M$ be a chain map. This means that u is a map $F_0 \rightarrow M$ such that $ud_1 = 0$, where d is the boundary operator in F . Since F_0 is projective, we can lift u to a map $\tau_0: F_0 \rightarrow F'_0 \otimes M$ such that $(\varepsilon' \otimes M)\tau_0 = u$. Recall that the boundary operator $d'_0: F'_0 \rightarrow F'_{-1}$ admits a factorization $d'_0 = \eta'\varepsilon'$, and consider $(d'_0 \otimes M)\tau_0 d_1$:

$$\begin{array}{ccc} F_1 & \xrightarrow{d_1} & F_0 \\ & \searrow \tau_0 & \swarrow u \\ & F'_0 \otimes M & \xrightarrow{\varepsilon' \otimes M} M \xleftarrow{\eta' \otimes M} F'_{-1} \otimes M \\ & \searrow d'_0 \otimes M & \uparrow \\ & & \end{array}$$

We have

$$\begin{aligned} (d'_0 \otimes M)\tau_0 d_1 &= (\eta' \otimes M)(\varepsilon' \otimes M)\tau_0 d_1 \\ &= (\eta' \otimes M)ud_1 \\ &= 0; \end{aligned}$$

so we may apply lemma 5.10 to conclude that τ_0 extends to a chain map $\tau: F \rightarrow F' \otimes M$. [Note that $F'_i \otimes M$ is relatively injective; indeed, it suffices to observe that if L is a free $\mathbb{Z}G$ -module then $L \otimes M$ is an induced module by

III.5.7.] This proves the surjectivity of $[F, F' \otimes M] \rightarrow [F, M]$. Now suppose $\tau: F \rightarrow F' \otimes M$ is a chain map such that $(\varepsilon' \otimes M)\tau: F \rightarrow M$ is null-homotopic, i.e., such that $(\varepsilon' \otimes M)\tau_0 = sd_0$ for some $s: F_{-1} \rightarrow M$. Lift s to a map $h_{-1}: F_{-1} \rightarrow F'_0 \otimes M$ satisfying $(\varepsilon' \otimes M)h_{-1} = s$, and consider $(d'_0 \otimes M)h_{-1}d_0$:

$$\begin{array}{ccccc}
 F_0 & \xrightarrow{d_0} & F_{-1} & & \\
 \downarrow \tau_0 & \nearrow h_{-1} & \downarrow s & & \\
 F'_0 \otimes M & \xrightarrow{\varepsilon' \otimes M} & M & \xrightarrow{\eta' \otimes M} & F'_{-1} \otimes M \\
 & \searrow d_0 \otimes M & & & \uparrow
 \end{array}$$

We have

$$\begin{aligned}
 (d'_0 \otimes M)h_{-1}d_0 &= (\eta' \otimes M)(\varepsilon' \otimes M)h_{-1}d_0 \\
 &= (\eta' \otimes M)sd_0 \\
 &= (\eta' \otimes M)(\varepsilon' \otimes M)\tau_0 \\
 &= (d'_0 \otimes M)\tau_0.
 \end{aligned}$$

As in the proof of the fundamental lemma I.7.4, this is precisely what is needed to extend h_{-1} to $(h_i)_{i \geq -1}$ satisfying $hd + (d' \otimes M)h = \tau$, cf. I.7.3b. We can now use Proposition 2.4b to extend $(h_i)_{i \geq -1}$ to a null-homotopy h of τ . Thus $[F, F' \otimes M] \rightarrow [F, M]$ is injective, whence (a). To prove (b), consider the dual $\bar{F} = \mathcal{H}om_G(F, \mathbb{Z}G)$. This is again projective and acyclic (cf. proof of 3.5), and we have a duality isomorphism $F \hat{\otimes}_G - \approx \mathcal{H}om_G(\bar{F}, -)$ by exercise 1 below. So (b) follows from (a). $\square$

Remark. It is perhaps surprising at first that we obtain the same conclusion here as in I.8.5 and I.8.6, since our map $\varepsilon' \otimes M$ is *not* a weak equivalence. On the other hand, we have a very strong hypothesis on F (acyclicity), and this compensates for the failure of $\varepsilon' \otimes M$ to be a weak equivalence.

It is now a routine matter to imitate the definitions in §V.4: Let $\varepsilon: F \rightarrow \mathbb{Z}$ and $\varepsilon': F' \rightarrow \mathbb{Z}$ be complete resolutions, and let $\varepsilon'': F'' \rightarrow \mathbb{Z}$ be either a complete resolution or the identity map $\mathbb{Z} \rightarrow \mathbb{Z}$. Then there are chain maps

$$\mathcal{H}om_G(F', F'' \otimes M) \otimes \mathcal{H}om_G(F, F' \otimes N) \rightarrow \mathcal{H}om_G(F, F'' \otimes M \otimes N)$$

and

$$\mathcal{H}om_G(F', F'' \otimes M) \otimes (F \hat{\otimes}_G (F' \otimes N)) \rightarrow F \hat{\otimes}_G (F'' \otimes M \otimes N),$$

defined as in V.4.2 and V.4.4. These induce *composition products*

$$\hat{H}^*(G, M) \otimes \hat{H}^*(G, N) \rightarrow \hat{H}^*(G, M \otimes N)$$

and

$$\hat{H}^*(G, M) \otimes \hat{H}_*(G, N) \rightarrow \hat{H}_*(G, M \otimes N).$$

The proof of V.4.6 now goes through, essentially verbatim, and yields:

(6.2) Theorem. *The composition products defined above coincide with the cup and cap products.*

EXERCISES

1. Let R be a ring, let C be a chain complex of finitely generated projective R -modules, and let $\bar{C}$ be the dual complex $\mathcal{H}om_R(C, R)$ of finitely generated projective right R -modules. Prove the following analogue of I.8.3b:

For any chain complex C' of left R -modules, there is a chain isomorphism $\varphi: \bar{C} \hat{\otimes}_R C' \xrightarrow{\sim} \mathcal{H}om_R(C, C')$. [Hint: Define $\varphi_{pq}: \bar{C}_p \otimes_R C'_q \rightarrow \text{Hom}_R(C_{-p}, C'_q)$ by $\langle \varphi_{pq}(u \otimes x'), x \rangle = (-1)^{pq} \langle u, x \rangle x'$ for $u \in \bar{C}_p = \text{Hom}_R(C_{-p}, R)$, $x' \in C'_q$, $x \in C_{-p}$. It is an isomorphism by I.8.3b. Now set $\varphi_n = \prod_{p+q=n} \varphi_{pq}: \prod_{p+q=n} \bar{C}_p \otimes_R C'_q \rightarrow \prod_{p+q=n} \text{Hom}_R(C_{-p}, C'_q) = \mathcal{H}om_R(C, C')_n$.] Similarly prove analogues of I.8.3c and I.8.3d.

2. Let C and $\bar{C}$ be as in exercise 1. For any $z \in (\bar{C} \hat{\otimes}_R C)_n$ and any chain complex C' , there is a graded map $\psi_z: \mathcal{H}om_R(C, C') \rightarrow \bar{C} \hat{\otimes}_R C'$ of degree n , given by $\psi_z(u) = (\text{id}_{\bar{C}} \hat{\otimes} u)(z)$. Show that there is a cycle $z \in (\bar{C} \hat{\otimes}_R C)_0$ such that for any C' , ψ_z is the inverse of the isomorphism $\varphi: \bar{C} \hat{\otimes}_R C' \xrightarrow{\sim} \mathcal{H}om_R(C, C')$ of exercise 1. [Hint: Let $z \in (\bar{C} \hat{\otimes}_R C)_0$ correspond to id_C under $\varphi: \bar{C} \hat{\otimes}_R C \xrightarrow{\sim} \mathcal{H}om_R(C, C)$. Then $z = (z_p)_{p \in \mathbb{Z}}$, where $z_p \in \bar{C}_{-p} \otimes_R C_p = (C_p)^* \otimes_R C_p$ corresponds to $(-1)^p \text{id}_{C_p}$ under the canonical isomorphism $(C_p)^* \otimes_R C_p \approx \text{Hom}_R(C_p, C_p)$. Checking the definition of ψ_z , you should find that ψ_z is induced by maps $\psi_{pq}: \text{Hom}_R(C_{-p}, C'_q) \rightarrow \bar{C}_p \otimes_R C'_q$ given by $\psi_{pq}(u) = (-1)^{(q+p)p} (\text{id}_{\bar{C}_p} \otimes u)(z_{-p})$. Exercise 7 of §I.8 now shows that $\psi_{pq} = \varphi_{pq}^{-1}$.]

- *3. Is the finiteness hypothesis in 6.1b necessary?

- *4. Let F be a complete resolution, let n be an arbitrary integer, and let Z be the module $Z_n(F)$ of n -cycles. Show that there is a ring homomorphism $\text{Hom}_G(Z, Z) \rightarrow [F, F]$ which is surjective and has as its kernel the group I of maps $Z \rightarrow Z$ which extend to maps $F_n \rightarrow Z$. Deduce that I is a 2-sided ideal in $\text{Hom}_G(Z, Z)$ and that there is a ring isomorphism $\text{Hom}_G(Z, Z)/I \approx \hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/|G| \cdot \mathbb{Z}$. [Hint: Use the techniques of the proof of 6.1.]

7 A Duality Theorem

It follows from the existence of finite type complete resolutions that the groups $\hat{H}^i(G, \mathbb{Z})$ are finitely generated. On the other hand, they are annihilated by $|G|$, so they are finite. The main purpose of this section is to show that $\hat{H}^i(G, \mathbb{Z})$ and $\hat{H}^{-i}(G, \mathbb{Z})$ are *dual* finite abelian groups; more precisely, we will show that the duality between them is given by the cup product

$$\hat{H}^i(G, \mathbb{Z}) \otimes \hat{H}^{-i}(G, \mathbb{Z}) \rightarrow \hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/|G| \cdot \mathbb{Z}.$$

We will give a proof which uses Tate homology theory and the cap product; a different proof, which uses only cup products and dimension-shifting, can be found in Cartan–Eilenberg [1956], XII.6. We begin by reviewing the relevant duality theory for abelian groups.

For any abelian group A we set $A' = \text{Hom}(A, \mathbb{Q}/\mathbb{Z})$. Since $\mathbb{Q}/\mathbb{Z}$ is injective, the (contravariant) functor $(\quad)'$ is *exact*. In case $nA = 0$ for some $n > 0$, we have $\text{Hom}(A, \mathbb{Q}/\mathbb{Z}) = \text{Hom}(A, n^{-1}\mathbb{Z}/\mathbb{Z}) \approx \text{Hom}(A, \mathbb{Z}/n\mathbb{Z})$; hence we can identify A' with $\text{Hom}(A, \mathbb{Z}_n)$. In particular, if A is cyclic of order n then so is A' . Consequently, $A' \approx A$ (non-canonically) for any finite abelian group A . As usual, we must pass to the double dual to get a canonical isomorphism: There is a natural map $A \rightarrow A''$ given by $a \mapsto (f \mapsto f(a))$, which is an isomorphism if A is finite.

A map $\rho: A \otimes B \rightarrow \mathbb{Q}/\mathbb{Z}$ gives rise to a map $\bar{\rho}: A \rightarrow B'$; we will say that ρ is a *duality pairing* if $\bar{\rho}$ is an isomorphism. Of course ρ also gives rise to a map $\bar{\rho}: B \rightarrow A'$, which is equal to the composite $B \rightarrow B'' \xrightarrow{\bar{\rho}} A'$. It follows easily that, if A and B are finite, ρ is a duality pairing if and only if $\bar{\rho}$ is an isomorphism. Thus a duality pairing between *finite* abelian groups provides an isomorphism of each one with the dual of the other. The canonical example of a duality pairing is, of course, the evaluation map $A' \otimes A \rightarrow \mathbb{Q}/\mathbb{Z}$, where A is arbitrary. Finally, if $nA = 0$ and $nB = 0$, then we can similarly speak of duality pairings $A \otimes B \rightarrow \mathbb{Z}_n$.

If G is a group and M is a G -module, then $M' = \text{Hom}(M, \mathbb{Q}/\mathbb{Z})$ inherits a G -action in the usual way: $(gu)(m) = u(g^{-1}m)$ for $g \in G, u \in M', m \in M$. There is an *evaluation pairing* $\rho: H^i(G, M') \otimes H_i(G, M) \rightarrow \mathbb{Q}/\mathbb{Z}$, obtained by composing the pairing $\langle \cdot, \cdot \rangle$ of §V.3 with the evaluation map $M' \otimes_G M \rightarrow \mathbb{Q}/\mathbb{Z}$. Similarly, if G is finite, there is a pairing $\hat{\rho}: \hat{H}^i(G, M') \otimes \hat{H}_i(G, M) \rightarrow \mathbb{Q}/\mathbb{Z}$.

(7.1) Proposition. *The pairings ρ and $\hat{\rho}$ are duality pairings. Thus $H^i(G, M') \approx H_i(G, M)$ for any G and M , and $\hat{H}^i(G, M') \approx \hat{H}_i(G, M)$ if G is finite.*

PROOF. Let F be a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Then $\mathcal{H}om_G(F, M') = \mathcal{H}om_G(F, \text{Hom}(M, \mathbb{Q}/\mathbb{Z})) \approx \mathcal{H}om_G(F \otimes M, \mathbb{Q}/\mathbb{Z}) = \mathcal{H}om(F \otimes_G M, \mathbb{Q}/\mathbb{Z}) = (F \otimes_G M)'$. Since $(\quad)'$ is exact, we can pass to homology to obtain $H^*(G, M') \approx H_*(G, M)'$. It is easy to check that this isomorphism corresponds to the pairing ρ . The argument for $\hat{\rho}$ is the same. $\square$

Assume now that G is finite. As we remarked briefly at the end of §4, there is an isomorphism $\hat{H}^i(G, M) \approx \hat{H}_{-1-i}(G, M)$. We wish to make this more precise by showing that there is an isomorphism of this type given by cap product with a “fundamental class”:

(7.2) Proposition. *There is an element $z \in \hat{H}_{-1}(G, \mathbb{Z})$ such that the cap product map $\cap z: \hat{H}^i(G, M) \rightarrow \hat{H}_{-1-i}(G, M)$ is an isomorphism for any G -module M .*

PROOF. Let (F, d) be a complete resolution of finite type, with $d_0 = \eta\varepsilon$ as in §3, and let $\bar{F}$ be the dual complex $\mathcal{H}om_G(F, \mathbb{Z}G)$. Thus $\bar{F}_i = (F_{-i})^*$. Then $\bar{F}$ is still

projective and acyclic (cf. proof of 3.5), and the boundary operator $\bar{F}_1 \rightarrow \bar{F}_0$ is the composite

$$(F_{-1})^* \xrightarrow{\eta^*} \mathbb{Z} \xrightarrow{\xi^*} (F_0)^*.$$

Except for indexing, then, $\bar{F}$ is a complete resolution; more precisely, $\bar{F}$ is the suspension ΣE of a complete resolution E . We now apply the duality isomorphism $\mathcal{H}om_G(F, M) \approx \bar{F} \otimes_G M$ of I.8.3b. This yields

$$\hat{H}^i(G, M) \approx H_{-i}(\bar{F} \otimes_G M) = H_{-i-1}(E \otimes_G M) = \hat{H}_{-i-1}(G, M).$$

I claim that this isomorphism is given, up to sign, by cap product with a universal element $z \in \hat{H}_{-1}(G, \mathbb{Z})$. There is an easy abstract proof of this, outlined in exercise 1 below. We will give here a direct but somewhat tedious proof, based on the composition product and the exercises of §6.

Let w_0 be the 0-cycle in $\bar{F} \hat{\otimes}_G F$ which corresponds to id_F under the canonical isomorphism $\bar{F} \hat{\otimes}_G F \approx \mathcal{H}om_G(F, F)$ of exercise 1 of §6. Using the notation and result of exercise 2 of §6, we find that the isomorphism

$$\mathcal{H}om_G(F, M) \xrightarrow{\sim} \bar{F} \otimes_G M$$

used above is the map ψ_{w_0} . Now $\bar{F} \hat{\otimes}_G F = \Sigma E \hat{\otimes}_G F$, so w_0 can also be regarded as a (-1) -cycle w_{-1} in $E \hat{\otimes}_G F$. Moreover, one discovers by checking definitions that $\psi_{w_{-1}}: \mathcal{H}om_G(F, M)_i \rightarrow (E \otimes_G M)_{i-1}$ is equal to $(-1)^i \psi_{w_0}: \mathcal{H}om_G(F, M)_i \rightarrow (\bar{F} \otimes_G M)_i$. Thus the isomorphism $\hat{H}^i(G, M) \rightarrow \hat{H}_{-i-1}(G, M)$ of the previous paragraph is induced, up to sign, by $\psi_{w_{-1}}$. On the other hand, $\psi_{w_{-1}}$ is precisely the composition product with w_{-1} as defined in §6; it therefore induces the cap product with the class $z \in \hat{H}_{-1}(G, \mathbb{Z})$ represented by w_{-1} . This proves the claim and the proposition. $\square$

Remark. The element z above is necessarily a generator of the cyclic group $\hat{H}_{-1}(G, \mathbb{Z}) = \mathbb{Z}/|G|\mathbb{Z}$; for the cap product isomorphism $\cap z: \hat{H}^0(G, \mathbb{Z}) \rightarrow \hat{H}_{-1}(G, \mathbb{Z})$ takes $1 \in \hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/|G|\mathbb{Z}$ to $1 \cap z = z$.

(7.3) Corollary. For any G -module M , the composite

$$\hat{H}^i(G, M') \otimes \hat{H}^{-1-i}(G, M) \xrightarrow{\sim} H^{-1}(G, M' \otimes M) \xrightarrow{\alpha_*} \hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z})$$

is a duality pairing, where α_* is induced by the canonical pairing $\alpha: M' \otimes M \rightarrow \mathbb{Q}/\mathbb{Z}$.

(Note that the statement makes sense because

$$\hat{H}^{-1}(G, \mathbb{Q}/\mathbb{Z}) = \ker\{\bar{N}: (\mathbb{Q}/\mathbb{Z})_G \rightarrow (\mathbb{Q}/\mathbb{Z})^G\} = |G|^{-1}\mathbb{Z}/\mathbb{Z}$$

and $\hat{H}^*(G, -)$ is annihilated by $|G|$.)

PROOF. By 7.1 and 7.2 we have $\hat{H}^i(G, M') \xrightarrow{\sim} \hat{H}_i(G, M') \xrightarrow{\sim} \hat{H}^{-1-i}(G, M')$. The composite isomorphism is given explicitly by $u \mapsto (v \mapsto \bar{\alpha}\langle u, v \cap z \rangle)$, where

$\bar{\alpha}: M' \otimes_G M \rightarrow \mathbb{Q}/\mathbb{Z}$ is induced by α . Since $\bar{\alpha}\langle u, v \cap z \rangle = \bar{\alpha}\langle u \cup v, z \rangle = \langle \alpha_*(u \cup v), z \rangle$, it follows that the composite

$$\hat{H}^i(G, M') \otimes \hat{H}^{-i}(G, M) \xrightarrow{\sim} \hat{H}^{-i}(G, M' \otimes M) \xrightarrow{\sim} \hat{H}^{-i}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\langle -, z \rangle} \mathbb{Q}/\mathbb{Z}$$

is a duality pairing. To complete the proof, we need only note that $\langle -, z \rangle$ maps $\hat{H}^{-i}(G, \mathbb{Q}/\mathbb{Z})$ isomorphically onto $|G|^{-i}\mathbb{Z}/\mathbb{Z}$. Indeed, $\langle -, z \rangle$ is the composite

$$\hat{H}^{-i}(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} \hat{H}_{-i}(G, \mathbb{Z}) \rightarrow \mathbb{Q}/\mathbb{Z},$$

where the first map is given by 7.1 and the second is evaluation at the generator $z \in \hat{H}_{-i}(G, \mathbb{Z})$. And it is obvious that if A is a finite cyclic group, then evaluation at a generator of A gives an isomorphism $A' \xrightarrow{\sim} |A|^{-1}\mathbb{Z}/\mathbb{Z}$. $\square$

We can now prove our main result:

(7.4) Theorem. *The cup product $\hat{H}^i(G, \mathbb{Z}) \otimes \hat{H}^{-i}(G, \mathbb{Z}) \rightarrow \hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/|G| \cdot \mathbb{Z}$ is a duality pairing.*

PROOF. Since $\hat{H}^*(G, \mathbb{Q}) = 0$, the coefficient sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ yields an isomorphism $\delta: \hat{H}^j(G, \mathbb{Q}/\mathbb{Z}) \xrightarrow{\sim} \hat{H}^{j+1}(G, \mathbb{Z})$ for all j . Moreover, we have (cf. V.3.3) a commutative diagram

$$\begin{array}{ccc} \hat{H}^{i-1}(G, \mathbb{Q}/\mathbb{Z}) \otimes \hat{H}^{-i}(G, \mathbb{Z}) & \xrightarrow{\cup} & \hat{H}^{-i}(G, \mathbb{Q}/\mathbb{Z}) \\ \delta \otimes \text{id} \downarrow \approx & & \downarrow \delta \approx \\ \hat{H}^i(G, \mathbb{Z}) \otimes \hat{H}^{-i}(G, \mathbb{Z}) & \xrightarrow{\cup} & \hat{H}^0(G, \mathbb{Z}). \end{array}$$

Since the top row is a duality pairing by 7.3 (with $M = \mathbb{Z}$), so is the bottom row. $\square$

Remark. The fact that $\hat{H}^i(G, \mathbb{Z})$ and $\hat{H}^{-i}(G, \mathbb{Z})$ are dual to one another can be proved quite easily by means of the universal coefficient theorem and the interpretation of $\hat{H}^*$ in terms of H^* and H_* . [Recall that $\text{Ext}(A, \mathbb{Z}) \approx A'$ if A is finite, cf. exercise 2 below.] What is not obvious, however, is that the cup product provides a duality pairing. This is the essential content of 7.4.

EXERCISES

1. The purpose of this exercise is to give an alternate proof that the isomorphism $\hat{H}^i(G, M) \xrightarrow{\sim} \hat{H}_{-i}(G, M)$ established in the first part of the proof of 7.2 is given, up to sign, by cap product with a fixed element $z \in H_{-i}(G, \mathbb{Z})$. Call that isomorphism ϕ and let $z = \phi(1)$.

(a) Show that ϕ is natural and that it is compatible with connecting homomorphisms in long exact sequences.

(b) Show that $\varphi: \hat{H}^0(G, M) \rightarrow \hat{H}_{-1}(G, M)$ is given by cap product with z . [By definition of z , φ and $\cap z$ agree on $1 \in \hat{H}^0(G, \mathbb{Z})$. If now M and $u \in \hat{H}^0(G, M)$ are arbitrary, there is a coefficient homomorphism $\mathbb{Z} \rightarrow M$ such that $1 \mapsto u$ under the induced map $\hat{H}^0(G, \mathbb{Z}) \rightarrow \hat{H}^0(G, M)$, so $\varphi(u) = u \cap z$ by naturality.]

(c) Now use dimension-shifting to show that φ and $\cap z$ agree in all dimensions, up to sign.

2. For any abelian torsion group A , prove that $\text{Ext}(A, \mathbb{Z}) \approx A'$. [Hint: Look at the long exact $\text{Ext}^*(A, -)$ -sequence associated to the short exact sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$. Equivalently, compute $\text{Ext}(A, \mathbb{Z})$ by means of the injective resolution $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ of $\mathbb{Z}$.]

3. If M is a G -module which is free as an abelian group, show that

$$\hat{H}^i(G, M^*) \otimes \hat{H}^{-i}(G, M) \xrightarrow{\sim} \hat{H}^0(G, M^* \otimes M) \rightarrow \hat{H}^0(G, \mathbb{Z})$$

is a duality pairing.

4. Let k be an arbitrary commutative ring and Q an injective k -module. Let $A' = \text{Hom}_k(A, Q)$ for any k -module A . If M is a kG -module, show that the pairing

$$\hat{H}^i(G, M') \otimes \hat{H}^{-1-i}(G, M) \xrightarrow{\sim} \hat{H}^{-1}(G, M' \otimes M) \rightarrow \hat{H}^{-1}(G, Q) \hookrightarrow Q$$

induces an isomorphism $\hat{H}^i(G, M') \xrightarrow{\sim} \hat{H}^{-1-i}(G, M)'$.

8 Cohomologically Trivial Modules

The theory to be presented in this section is due to Nakayama and Rim. Our treatment is based on Serre [1968].

A G -module M (where G is finite) is said to be *cohomologically trivial* if $\hat{H}^i(H, M) = 0$ for all $i \in \mathbb{Z}$ and all $H \subseteq G$. For example, any induced module $\mathbb{Z}G \otimes A$ is cohomologically trivial by 5.3, since $\mathbb{Z}G \otimes A$ is still induced when regarded as an H -module for any $H \subseteq G$. [This is an easy special case of the double-coset formula III.5.6b.] In particular, if k is an arbitrary commutative ring, it follows that any free kG -module is cohomologically trivial and hence that any projective kG -module is cohomologically trivial. Our goals in this section are (a) to prove that, conversely, under suitable hypotheses cohomologically trivial kG -modules are projective; and (b) to find simple criteria for checking that a module is cohomologically trivial. These results have applications to class field theory (cf. Serre [1968]) as well as to algebraic K -theory and homotopy theory.

We begin with the case where G is a p -group for some prime p . In this case everything will be deduced from the following simple result:

(8.1) Proposition. *If G is a p -group and M is a G -module in which every element has order a power of p , then $M^G \neq 0$.*

PROOF. We may assume that M is finitely generated as a G -module, hence also as an abelian group. But then M is finite, of order a power of p . Since every G -orbit in M has order p^a for some $a \geq 0$, it follows that $|M^G| \equiv |M| \equiv 0 \pmod{p}$, so that $M^G \neq 0$. $\square$

(8.2) Corollary. *If k is a field of characteristic p , G is a p -group, and M is a simple kG -module, then $M \approx k$, with trivial G -action.*

(Recall that a *simple* module is a non-zero module which contains no proper non-zero submodule.)

PROOF. Since $M^G \neq 0$ by 8.1, we must have $M^G = M$, so G acts trivially on M . Simplicity now implies that $\dim_k M = 1$. $\square$

Let I be the *augmentation ideal* of kG , i.e., the kernel of the k -algebra homomorphism $\varepsilon: kG \rightarrow k$ such that $\varepsilon(g) = 1$ for all $g \in G$. We can also describe I as the annihilator of the kG -module k .

(8.3) Corollary. *If G is a p -group and k is a field of characteristic p , then the augmentation ideal I of kG is nilpotent.*

PROOF. Choose a composition series $0 = J_0 \subset J_1 \subset \cdots \subset J_n = kG$ for kG , regarded as a left kG -module. Thus each J_i is a left ideal and J_i/J_{i-1} is simple for $1 \leq i \leq n$. [The existence of a composition series follows from the finite-dimensionality of kG over k .] By 8.2, I annihilates J_i/J_{i-1} , hence $IJ_i \subseteq J_{i-1}$. Consequently, $I^n = I^n J_n \subseteq J_0 = 0$. $\square$

If G is cyclic of order $q = p^a$, for example, we can easily see directly that I is nilpotent; for I is generated by $t - 1$ where t is a generator of G (cf. §I.2, exercise 1b), and $(t - 1)^q = t^q - 1 = 0$ since $\text{char } k = p$.

(8.4) Corollary. *Let G be a p -group and k a field of characteristic p . If M is a kG -module such that $H_0(G, M) = 0$, then $M = 0$.*

PROOF. It is easy to see that $H_0(G, M) = M/IM$ (cf. first paragraph of §II.2). Thus $H_0(G, M) = 0 \Rightarrow M = IM \Rightarrow M = I^n M$ for all n . In view of 8.3, this implies that $M = 0$. $\square$

We can now achieve, for kG -modules as above, the two goals (a) and (b) stated at the beginning of this section:

(8.5) Theorem. *Let G be a p -group and k a field of characteristic p . The following conditions on a kG -module M are equivalent:*

- (i) M is free.
- (ii) M is projective.
- (iii) M is cohomologically trivial.
- (iv) $H_1(G, M) = 0$.
- (v) $\hat{H}^i(G, M) = 0$ for some $i \in \mathbb{Z}$.

PROOF. Clearly (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (v). To complete the proof we will show (iv) $\Rightarrow$ (i) and (v) $\Rightarrow$ (iv). Choose a k -basis for the vector space M_G , and lift the basis elements to elements m_j ($j \in J$) in M . Let F be a free kG -module with basis $(e_j)_{j \in J}$ and define $f: F \rightarrow M$ by $f(e_j) = m_j$. By construction, then, $H_0(G, f): H_0(G, F) \rightarrow H_0(G, M)$ is an isomorphism. By right exactness of $H_0(G, -)$, it follows that $H_0(G, \text{coker } f) = 0$, and 8.4 now implies that $\text{coker } f = 0$. Consider now the long exact homology sequence associated to $0 \rightarrow \ker f \rightarrow F \rightarrow M \rightarrow 0$. If $H_1(G, M) = 0$, then the fact that $H_0(G, f)$ is an isomorphism implies that $H_0(G, \ker f) = 0$. Applying 8.4 again, we conclude that $\ker f = 0$, so f is an isomorphism. Thus M is free and we have proven (iv) $\Rightarrow$ (i).

Assume now that (v) holds. By dimension-shifting we can find a kG -module N such that $\hat{H}^n(G, N) \approx \hat{H}^{n+i+2}(G, M)$ for all $n \in \mathbb{Z}$. [Note that the dimension-shifting argument does not take us out of the category of kG -modules, because $\mathbb{Z}G \otimes A$ is a kG -module if A is a k -module.] In particular, $H_1(G, N) = \hat{H}^{-2}(G, N) = \hat{H}^i(G, M) = 0$, so N is free by the previous paragraph. But then $\hat{H}^*(G, N) = 0$, hence $\hat{H}^*(G, M) = 0$, hence $H_1(G, M) = 0$. Thus (v) $\Rightarrow$ (iv). $\square$

Note that the theorem implies, in particular, that $H^i(G, \mathbb{Z}_p) \neq 0$ for all $i > 0$ if G is a (non-trivial) p -group.

Remarks

1. Corollary 8.2 can be interpreted as saying that I is the Jacobson radical of kG . From this point of view, 8.3 is simply the well-known result that the Jacobson radical of an Artin ring is nilpotent, and 8.4 is a special case of "Nakayama's lemma." The equivalence of (i), (ii), and (iv) in 8.5 could also be stated in a more general context, as a theorem about modules over a local ring. [It follows from 8.3 that kG is a (non-commutative) local ring, with I as its unique maximal ideal.]

2. The proof above yields the following more precise version of the implication (iv) $\Rightarrow$ (i): For any kG -module M , $\dim_k H_0(G, M)$ is the minimal number of generators of M and $\dim_k H_1(G, M)$ is the minimal number of defining relations among any minimal set of generators.

Next we will consider more general G -modules M , still assuming that G is a p -group. Let $M_p = M \otimes \mathbb{Z}_p = M/pM$.

(8.6) Lemma. *Let G be a p -group and M a G -module which is p -torsion-free. For any $i \in \mathbb{Z}$, $\hat{H}^i(G, M_p) = 0$ if and only if $\hat{H}^i(G, M) = \hat{H}^{i+1}(G, M) = 0$.*

PROOF. From the short exact sequence $0 \rightarrow M \xrightarrow{p} M \rightarrow M_p \rightarrow 0$ we obtain an exact sequence

$$\hat{H}^i(G, M) \xrightarrow{p} \hat{H}^i(G, M) \rightarrow \hat{H}^i(G, M_p) \rightarrow \hat{H}^{i+1}(G, M) \xrightarrow{p} \hat{H}^{i+1}(G, M).$$

If $\hat{H}^i(G, M) = \hat{H}^{i+1}(G, M) = 0$, then clearly this implies that $\hat{H}^i(G, M_p) = 0$. Conversely, if $\hat{H}^i(G, M_p) = 0$, then we conclude that $\hat{H}^i(G, M)$ is p -divisible and $\hat{H}^{i+1}(G, M)$ is p -torsion-free. But $\hat{H}^*(G, M)$ is annihilated by $|G|$, which is a power of p , so $\hat{H}^i(G, M) = \hat{H}^{i+1}(G, M) = 0$. $\square$

(8.7) Theorem. *If G is a p -group, then the following two conditions are equivalent for any G -module M :*

- (i) M is cohomologically trivial.
- (ii) $\hat{H}^i(G, M) = 0$ for two consecutive integers i .

If M is p -torsion-free, then (i) and (ii) are also equivalent to:

- (iii) $\hat{H}^i(G, M_p) = 0$ for some i .
- (iv) M_p is cohomologically trivial.
- (v) M_p is free over $\mathbb{Z}_p[G]$.

PROOF. Suppose first that M is p -torsion-free. Then (i) $\Rightarrow$ (ii) trivially; (ii) $\Rightarrow$ (iii) by 8.6; (iii) $\Leftrightarrow$ (iv) $\Leftrightarrow$ (v) by 8.5; and (iv) $\Rightarrow$ (i) by 8.6 applied to G and all of its subgroups. If M is now arbitrary, then we can prove the equivalence of (i) and (ii) by using dimension-shifting to reduce to the previous case; for we can choose an exact sequence $0 \rightarrow M' \rightarrow F \rightarrow M \rightarrow 0$ with F free over $\mathbb{Z}G$ and then apply the previous case to M' . $\square$

An important consequence of 8.7 is that cohomological triviality can be checked (if G is a p -group) by looking only at $\hat{H}^*(G, M)$, even though the definition involves $\hat{H}^*(H, M)$ for arbitrary subgroups $H \subseteq G$.

We turn now to the case where G is an arbitrary finite group. For each prime p dividing $|G|$, choose a p -Sylow subgroup $G(p)$.

(8.8) Proposition. *A G -module M is cohomologically trivial if and only if its restriction to $G(p)$ is cohomologically trivial for each p .*

PROOF. The “only if” part is trivial. To prove the “if” part, note first that if the restriction of M to $G(p)$ is cohomologically trivial, then $\hat{H}^*(P, M) = 0$ for any p -group $P \subseteq G$. For we know that $gPg^{-1} \subseteq G(p)$ for some $g \in G$, so there is a conjugation isomorphism $\hat{H}^*(P, M) \approx \hat{H}^*(gPg^{-1}, M) = 0$. If $H \subseteq G$ is now an arbitrary subgroup, then we know from transfer theory (cf. §III.10) that $\hat{H}^*(H, M) = \bigoplus_p \hat{H}^*(H, M)_{(p)} \hookrightarrow \bigoplus_p \hat{H}^*(H(p), M) = 0$, where p ranges over the primes dividing $|H|$ and $H(p)$ is a p -Sylow subgroup of H . Thus M is cohomologically trivial. $\square$

Combining 8.8 with our previous results on cohomological triviality over p -groups, we obtain a solution to problem (b) stated at the beginning of the section. In particular, we have:

(8.9) Theorem. *A G -module M is cohomologically trivial if and only if for each prime p there are two consecutive integers i such that $\hat{H}^i(G(p), M) = 0$.*

The solution to problem (a) (with $k = \mathbb{Z}$) is given by the following theorem of Rim:

(8.10) Theorem. *If M is a G -module which is $\mathbb{Z}$ -free and cohomologically trivial, then M is $\mathbb{Z}G$ -projective.*

PROOF. Choose a short exact sequence $0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0$ with F $\mathbb{Z}G$ -free. We will prove that this sequence splits, so that M is a direct summand of F and hence is projective. I claim that the obstruction to splitting the sequence lies in $H^1(G, \text{Hom}(M, K))$. More precisely, we have a short exact sequence of G -modules $0 \rightarrow \text{Hom}(M, K) \rightarrow \text{Hom}(M, F) \rightarrow \text{Hom}(M, M) \rightarrow 0$ since M is $\mathbb{Z}$ -free, and this yields an exact sequence $\text{Hom}_G(M, F) \rightarrow \text{Hom}_G(M, M) \xrightarrow{\delta} H^1(G, \text{Hom}(M, K))$. [Recall that

$$\text{Hom}_G(-, -) = \text{Hom}(-, -)^G = H^0(G, \text{Hom}(-, -)).]$$

Hence the extension splits if and only if $\delta(\text{id}_M) = 0$. It will therefore suffice to prove:

(8.11) Lemma. *Let M and K be G -modules which are $\mathbb{Z}$ -free. If M is cohomologically trivial then so is $\text{Hom}(M, K)$.*

PROOF. By 8.8 we need only consider the case where G is a p -group, in which case it suffices by 8.7 to prove that $\text{Hom}(M, K)_p$ is cohomologically trivial. From the exact sequence $0 \rightarrow K \xrightarrow{p} K \rightarrow K_p \rightarrow 0$ we get (since M is $\mathbb{Z}$ -free) an exact sequence $0 \rightarrow \text{Hom}(M, K) \xrightarrow{p} \text{Hom}(M, K) \rightarrow \text{Hom}(M, K_p) \rightarrow 0$. Thus $\text{Hom}(M, K)_p \approx \text{Hom}(M, K_p) = \text{Hom}(M_p, K_p)$. But M_p is $\mathbb{Z}_p[G]$ -free by 8.7, so $\text{Hom}(M_p, K_p)$ is induced (cf. §III.5, exercise 2b) and hence cohomologically trivial. $\square$

Remark. The group $H^1(G, \text{Hom}(M, K))$ which arose in the proof of 8.10 is isomorphic to $\text{Ext}_{\mathbb{Z}G}^1(M, K)$ by III.2.2. The reader familiar with the theory of extensions of modules will therefore not be surprised that this group contains the obstruction to splitting the extension $0 \rightarrow K \rightarrow F \rightarrow M \rightarrow 0$.

Finally, we will deduce from 8.10 a characterization (also due to Rim) of cohomologically trivial modules which are not necessarily $\mathbb{Z}$ -free. If R is a ring and M is an R -module, the *projective dimension* of M , denoted $\text{proj dim } M$ or $\text{proj dim}_R M$, is defined to be the infimum of the set of integers n such that M admits a projective resolution $0 \rightarrow P_n \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$ of length n . (In particular, $\text{proj dim } M = \infty$ if M does not admit a projective resolution of finite length.)

(8.12) Theorem. *The following conditions on a $\mathbb{Z}G$ -module M are equivalent:*

- (i) M is cohomologically trivial.
- (ii) $\text{proj dim } M \leq 1$.
- (iii) $\text{proj dim } M < \infty$.

PROOF. (i) $\Rightarrow$ (ii): Suppose M is cohomologically trivial, and choose a short exact sequence $0 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$ with P_0 $\mathbb{Z}G$ -projective. Then P_1 is cohomologically trivial and $\mathbb{Z}$ -free, so P_1 is projective by 8.10 and (ii) holds. (ii) $\Rightarrow$ (iii) trivially. (iii) $\Rightarrow$ (i): Suppose $\text{proj dim } M < \infty$ and let $0 \rightarrow P_n \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$ be a projective resolution of finite length. Breaking up this resolution into short exact sequences $0 \rightarrow Z_i \rightarrow P_i \rightarrow Z_{i-1} \rightarrow 0$, one sees by descending induction on i that Z_i is cohomologically trivial. In particular, $M = Z_{-1}$ is cohomologically trivial. $\square$

Rim's results can be used to provide examples of projective modules over $\mathbb{Z}G$. For example, suppose $J \subset \mathbb{Z}G$ is a left ideal of finite index m , where m is relatively prime to $|G|$. Then $|G|$ is invertible in $M = \mathbb{Z}G/J$, so M is cohomologically trivial. It now follows as in the proof of (i) $\Rightarrow$ (ii) above that J is projective. (See also exercise 7 of §VIII.2 for an alternative proof that $\text{proj dim } M \leq 1$ when $|G|$ is invertible in M .)

EXERCISES

1. Give an example where $\hat{H}^*(G, M) = 0$ but M is not cohomologically trivial. [Hint: Take G to be cyclic.]
2. Prove the following improvement of 8.11: If M is cohomologically trivial and $\mathbb{Z}$ -free, then $\text{Hom}(M, K)$ is cohomologically trivial for any G -module K . [Use 8.10. Alternatively, choose an exact sequence $0 \rightarrow L \rightarrow F \rightarrow K \rightarrow 0$ with F free, and apply 8.11 to $\text{Hom}(M, L)$ and $\text{Hom}(M, F)$.]
3. (a) If M and P are $\mathbb{Z}G$ -modules such that M is $\mathbb{Z}$ -free and P is $\mathbb{Z}G$ -projective, show that any exact sequence $0 \rightarrow P \rightarrow E \rightarrow M \rightarrow 0$ splits. [Hint: Argue as in the proof of 8.10, or, more simply, use the fact that P is relatively injective.]
 (b) Using (a), give a direct proof (without using the results of this section) that (iii) $\Rightarrow$ (ii) in 8.12.
4. Let G be a group such that there exists a free, finite G -CW-complex X with $H_*(X) \approx H_*(S^{2k-1})$. Prove that G admits a complete resolution which is periodic of period $2k$. [Hint: If B is the module of $(2k-1)$ -boundaries of $C_*(X)$, then $C_{2k-1}(X)/B$ is $\mathbb{Z}$ -free and has finite projective dimension.]

9 Groups with Periodic Cohomology

A finite group G is said to have *periodic cohomology* if for some $d \neq 0$ there is an element $u \in \hat{H}^d(G, \mathbb{Z})$ which is invertible in the ring $\hat{H}^*(G, \mathbb{Z})$. Cup product with u then gives a *periodicity isomorphism*

$$u \cup - : \hat{H}^n(G, M) \xrightarrow{\sim} \hat{H}^{n+d}(G, M)$$

for all $n \in \mathbb{Z}$ and all G -modules M . In particular, taking $n = 0$ and $M = \mathbb{Z}$, we see that $\hat{H}^d(G, \mathbb{Z}) \approx \mathbb{Z}/|G| \cdot \mathbb{Z}$ and that u generates $\hat{H}^d(G, \mathbb{Z})$.

If we know that a group G has periodic cohomology, then the task of computing $\hat{H}^*(G)$ is obviously enormously simplified. It is therefore of interest to find criteria for deciding whether G has periodic cohomology, and that is what we will do in this section.

(9.1) Theorem. *The following conditions are equivalent:*

- (i) G has periodic cohomology.
- (ii) There exist integers n and d , with $d \neq 0$, such that $\hat{H}^n(G, M) \approx \hat{H}^{n+d}(G, M)$ for all G -modules M .
- (iii) For some $d \neq 0$, $\hat{H}^d(G, \mathbb{Z}) \approx \mathbb{Z}/|G| \cdot \mathbb{Z}$.
- (iv) For some $d \neq 0$, $\hat{H}^d(G, \mathbb{Z})$ contains an element u of order $|G|$.

PROOF. (i) $\Rightarrow$ (ii) trivially. If (ii) holds then we have $\hat{H}^n(G, M) \approx \hat{H}^{n+d}(G, M)$ for all $n \in \mathbb{Z}$ by dimension-shifting; taking $n = 0$ and $M = \mathbb{Z}$, we obtain (iii). (iii) $\Rightarrow$ (iv) trivially. Finally, suppose (iv) holds. Then there is a map $\hat{H}^d(G, \mathbb{Z}) \rightarrow \mathbb{Q}/\mathbb{Z}$ such that $u \mapsto 1/|G| \bmod \mathbb{Z}$ (cf. proof of III.4.3). By the duality theorem 7.4, this map is given by

$$\hat{H}^d(G, \mathbb{Z}) \xrightarrow{-\cup v} \hat{H}^0(G, \mathbb{Z}) \approx |G|^{-1}\mathbb{Z}/\mathbb{Z} \hookrightarrow \mathbb{Q}/\mathbb{Z}$$

for some $v \in \hat{H}^{-d}(G, \mathbb{Z})$. Thus $uv = 1$, whence (i). □

For example, if G acts freely on S^{2k-1} as in §I.6, then G obviously admits a periodic complete resolution of period $d = 2k$, so 9.1(ii) holds and G has periodic cohomology. [Alternatively, one can show directly, without using 9.1, that $\hat{H}^d(G, \mathbb{Z})$ contains an invertible element; cf. §V.3, exercise 3.]

One way to obtain examples of this is to start with a linear action of a finite group G on an even-dimensional real vector space V , such that G acts freely on $V - \{0\}$ (i.e., such that 1 is not an eigenvalue of any non-trivial element $g \in G$); such an action is called a *fixed-point-free representation* of G . By choosing a G -invariant sphere $S \subset V - \{0\}$ (e.g., the unit sphere relative to a G -invariant inner product on V), we obtain a free action of G on an odd-dimensional sphere. In order to apply §I.6, of course, we must verify that S can be chosen to have a G -equivariant CW -structure. This follows from a general triangulation theorem, but in the present case there is a much more elementary argument, due to Illman [1978]: Let $(e_i)_{1 \leq i \leq 2k}$ be a basis for V and let C be the convex hull of the finite set $\{\pm g e_i; g \in G, 1 \leq i \leq 2k\}$. Then C is a convex cell containing 0 as an interior point, hence its boundary S is a topological sphere in $V - \{0\}$, with a CW -structure given by the natural (rectilinear) faces of C (see, for example, Hudson [1969], Chapter I). Since G acts linearly on V and maps C into itself, it is clear that S is a G -complex.

Remark. The question of the existence of an equivariant CW -structure is actually a red herring. For one can show, without assuming an equivariant

CW-structure, that if G acts freely on S^{2k-1} then $\hat{H}^*(G)$ is periodic of period $2k$. A proof will be outlined in exercise 4c of §VII.7.

We can now give some specific examples of groups with periodic cohomology:

(9.2) EXAMPLES

1. Any finite cyclic group G admits a 2-dimensional fixed-point-free representation as a group of rotations, hence $\hat{H}^*(G)$ is periodic of period 2.

2. Let $\mathbb{H}$ be the quaternion algebra $\mathbb{R} \oplus \mathbb{R}i \oplus \mathbb{R}j \oplus \mathbb{R}k$. If G is a finite subgroup of the multiplicative group $\mathbb{H}^*$, then the multiplication action of G on $\mathbb{H}$ yields a 4-dimensional fixed-point-free representation of G , since $\mathbb{H}$ is a division algebra. Thus $\hat{H}^*(G)$ is periodic of period 4. The most obvious examples of finite subgroups of $\mathbb{H}^*$ are the generalized quaternion groups Q_{4m} ($m \geq 2$) which we studied in §IV.4. In addition to these (and, of course, the cyclic groups), there are precisely three finite subgroups of $\mathbb{H}^*$, up to conjugacy (cf. Wolf [1974], §2.6): the binary tetrahedral group (of order 24), the binary octahedral group (of order 48), and the binary icosahedral group (of order 120). The latter group G is particularly interesting because it is a *perfect* group, i.e., $G_{ab} = 0$. Using Poincaré duality, one can deduce from this that the quotient manifold S^3/G has the same homology as S^3 . This was discovered by Poincaré [1904]; it was the first example of a 3-manifold homologically equivalent to S^3 but not homeomorphic to S^3 .² Another interesting fact about the binary icosahedral group G is given in the following surprising theorem of Zassenhaus (cf. Wolf [1974], 6.2): Up to isomorphism, G is the unique perfect group which admits a fixed-point-free representation.

3. Let m, n , and r be positive integers such that $r^n \equiv 1 \pmod{m}$. Assume that the following two conditions hold: (a) m and n are relatively prime; (b) if k is the multiplicative order of $r \pmod{m}$ (so necessarily $k|n$), then every prime divisor of n divides n/k . Let G be the semi-direct product $\mathbb{Z}_m \rtimes \mathbb{Z}_n$, where the generator of $\mathbb{Z}_n$ acts on $\mathbb{Z}_m$ by multiplication by r . I claim that G admits a $2k$ -dimensional fixed-point-free representation and hence $\hat{H}^*(G)$ is periodic of period $2k$. To see this, note first that the subgroup $A \subset \mathbb{Z}_n$ of index k acts trivially on $\mathbb{Z}_m$, so G contains $C = \mathbb{Z}_m \times A$, which is cyclic by (a). From (b) we see that A contains every subgroup of $\mathbb{Z}_n$ of prime order, and it follows easily that C contains every subgroup of G of prime order. Choose a fixed-point-free representation of C on a 2-dimensional vector space W , and form the induced

² Poincaré had asked in an earlier paper whether such a 3-manifold could exist. In view of this example, Poincaré reformulated his question, adding the condition that the manifold be simply-connected. This question, or rather the assertion that there can be no such manifold, has since become known as the Poincaré conjecture.

module $V = \mathbb{Z}G \otimes_{\mathbb{Z}C} W = \mathbb{R}G \otimes_{\mathbb{R}C} W$. Since $C \triangleleft G$, we have $\text{Res}_C^G V = \bigoplus_{g \in G/C} gW$ by III.5.6b, and each gW is clearly a fixed-point-free representation of C . Consequently, V is a fixed-point-free representation of C . But then V is also fixed-point-free as a representation of G , for a non-trivial isotropy group G_v ($v \in V - \{0\}$) would contain a subgroup of prime order and hence would meet C nontrivially.

See Wolf [1974] for a complete classification of groups which admit a fixed-point-free representation.

Remarks

1. If G has periodic cohomology then so does any subgroup H . This follows from the fact that the restriction map $\hat{H}^*(G, \mathbb{Z}) \rightarrow \hat{H}^*(H, \mathbb{Z})$, being a ring homomorphism, must take invertible elements to invertible elements. [Alternatively, use Shapiro's lemma.]

2. If G is abelian but not cyclic, then G does *not* have periodic cohomology. For G must contain a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$ for some prime p , and direct calculation (using the Künneth formula, for example) shows that $\hat{H}^n(\mathbb{Z}_p \times \mathbb{Z}_p; \mathbb{Z}_p)$ has $\mathbb{Z}_p$ -dimension $n + 1$ for $n \geq 0$ and hence is not periodic.

We return now to the general question of finding criteria for a group to have periodic cohomology. In case G is a p -group, the question is completely settled by the following result. Recall that an *elementary abelian p -group* of rank $r \geq 0$ is a group isomorphic to $\mathbb{Z}_p^r = \mathbb{Z}_p \times \cdots \times \mathbb{Z}_p$ (r factors).

(9.3) Proposition. *If G is a p -group for some prime p , then the following conditions are equivalent:*

- (i) G has periodic cohomology.
- (ii) Every abelian subgroup of G is cyclic.
- (iii) Every elementary abelian p -subgroup of G has rank ≤ 1 .
- (iv) G has a unique subgroup of order p .
- (v) G is a cyclic or generalized quaternion group.

(The generalized quaternion groups, of course, can only occur if $p = 2$.)

PROOF. We have (v) $\Rightarrow$ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) by the examples and remarks above. (iii) $\Rightarrow$ (iv): Since G is a p -group, one knows that G contains a central subgroup of order p (cf. Hall [1959], Theorem 4.3.1). If there were any other subgroup of order p , the two would generate an elementary abelian p -group of rank 2, contrary to (iii). Finally, the implication (iv) $\Rightarrow$ (v) is Theorem IV.4.3. $\square$

To pass from the p -group case to the general case, we have:

(9.4) Proposition. *A finite group G has periodic cohomology if and only if every Sylow subgroup has periodic cohomology.*

PROOF. The “only if” part follows from Remark 1 above. To prove the “if” part, fix a prime p and let $H \subseteq G$ be a p -Sylow subgroup. By hypothesis, $\hat{H}^*(H, \mathbb{Z})$ contains an invertible element u of degree $d \neq 0$, and I claim that some power of u is G -invariant in the sense of §III.10. In fact, let $e > 0$ be an integer such that $a^e = 1$ for all $a \in (\mathbb{Z}/|H| \cdot \mathbb{Z})^*$, the latter being the group of units in $\mathbb{Z}/|H| \cdot \mathbb{Z}$. For any $g \in G$, the elements $v = \text{res}_{H \cap gHg^{-1}}^H u$ and $w = \text{res}_{H \cap gHg^{-1}}^{gHg^{-1}} gu$ are invertible in $\hat{H}^*(H \cap gHg^{-1}, \mathbb{Z})$ and hence generate the cyclic group $\hat{H}^d(H \cap gHg^{-1}, \mathbb{Z})$. We therefore have $v = aw$ for some $a \in (\mathbb{Z}/|H| \cdot \mathbb{Z})^*$, since the map $(\mathbb{Z}/|H| \cdot \mathbb{Z})^* \rightarrow (\mathbb{Z}/|H'| \cdot \mathbb{Z})^*$ is surjective for any $H' \subseteq H$. Thus $v^e = w^e$, and it follows that $u^e \in \hat{H}^{de}(H, \mathbb{Z})$ is invariant. Similarly, u^{-e} is invariant. In view of the Tate cohomology version of Theorem III.10.3, this proves that the ring $\hat{H}^*(G, \mathbb{Z})_{(p)}$ has an invertible element $u(p)$ of non-zero degree $d(p)$. Finally, replacing $u(p)$ by a suitable power $u(p)^{e(p)}$, we may assume that $d(p)$ is the same for all primes $p \mid |G|$; since $\hat{H}^*(G, \mathbb{Z})$ is the direct product of the rings $\hat{H}^*(G, \mathbb{Z})_{(p)}$ (cf. exercise 2 of §5), it follows that $\hat{H}^*(G, \mathbb{Z})$ contains an invertible element of positive degree. $\square$

Combining 9.3 and 9.4, we have:

(9.5) Theorem. *The following conditions are equivalent for a finite group G :*

- (i) G has periodic cohomology.
- (ii) Every abelian subgroup of G is cyclic.
- (iii) Every elementary abelian p -subgroup of G (p prime) has rank ≤ 1 .
- (iv) The Sylow subgroups of G are cyclic or generalized quaternion groups.

Using the criterion (iv) of this theorem, we can now give some examples of groups with periodic cohomology, beyond those of 9.2.

(9.6) EXAMPLES

1. If m and n are relatively prime integers, then any semi-direct product $\mathbb{Z}_m \rtimes \mathbb{Z}_n$ (arbitrary action) has cyclic Sylow subgroups and hence has periodic cohomology. [Conversely, a theorem of Burnside (cf. Wolf [1974], 5.4) implies that any group with cyclic Sylow subgroups is a semi-direct product of this type.] This generalizes example 3 of 9.2.

2. Let p be a prime and let $G = SL_2(\mathbb{F}_p)$, the group of 2×2 matrices of determinant 1 over the prime field $\mathbb{F}_p$. Then G has periodic cohomology. To see this, note first that $|G| = p(p^2 - 1)$. [There are $p^2 - 1$ possibilities for the first column of a matrix in G ; given the first column, the number of possibilities for the second is $(p^2 - p)/(p - 1) = p$.] Thus a p -Sylow subgroup of G is of order p and hence is cyclic. Now let l be a prime $\neq p$. Then the polynomial $x^l - 1$ is separable over $\mathbb{F}_p$, so a matrix of order l is diagonalizable over the finite extension k obtained from $\mathbb{F}_p$ by adjoining the l -th roots of unity. Two

commuting elements of G of order l are then simultaneously diagonalizable and hence generate a group isomorphic to

$$\left\{ \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} : \lambda \in k, \lambda^l = 1 \right\},$$

which is of order l . Thus G does not contain a subgroup isomorphic to $\mathbb{Z}_l \times \mathbb{Z}_l$ and hence G has periodic cohomology. [See exercise 8 below for a different proof, which consists of explicitly exhibiting cyclic or quaternionic Sylow subgroups.]

Remarks

1. The groups $SL_2(\mathbb{F}_p)$ for $p = 2, 3, 5$ have occurred earlier in our examples. Namely, $SL_2(\mathbb{F}_2) \approx \mathbb{Z}_3 \rtimes \mathbb{Z}_2$, $SL_2(\mathbb{F}_3)$ is isomorphic to the binary tetrahedral group, and $SL_2(\mathbb{F}_5)$ is isomorphic to the binary icosahedral group.

2. A complete classification of the groups with periodic cohomology has been obtained by Suzuki [1955], extending earlier work of Zassenhaus. (See also Wolf [1974], 6.1 and 6.3.) Not all of these groups can act freely on a sphere. In fact, Milnor [1957a] showed that a group which acts freely on a sphere has at most one element of order 2. (Thus, for example, the dihedral group D_{2n} for n odd cannot act freely on a sphere, although all of its Sylow subgroups are cyclic.) Conversely, Madsen, Thomas, and Wall [1976] have proven that if G has periodic cohomology and at most one element of order 2 then G acts freely on a sphere. This applies, for example, to the groups $SL_2(\mathbb{F}_p)$ (p odd) since

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

is the only element of order 2. These groups do not, however, admit a free *orthogonal* action on a sphere (i.e., a fixed-point-free representation), except for $p = 3$ and $p = 5$.

The condition that *all* Sylow subgroups of a group G be cyclic or generalized quaternion is obviously very restrictive. What is much more common, however, is for there to be at least one prime p such that the p -Sylow subgroups are cyclic or generalized quaternion. It is therefore of interest to observe that the results of this section can be “localized” at a single prime p . Thus we say that G has *p-periodic cohomology* if the ring $\hat{H}^*(G, \mathbb{Z})_{(p)}$ contains an invertible element of non-zero degree d . Cup product with such an element then gives periodicity isomorphisms $\hat{H}^n(G, M)_{(p)} \approx \hat{H}^{n+d}(G, M)_{(p)}$ for any G -module M . The reader can easily state and prove analogues of 9.1, 9.4, and 9.5 for this situation. In particular, one can prove:

(9.7) Theorem. *For any finite group G and prime p dividing $|G|$, the following conditions are equivalent:*

- (i) G has p -periodic cohomology.
- (ii) The ring $\hat{H}^*(G, \mathbb{Z}_p)$ contains an invertible element of non-zero degree.
- (iii) Every elementary abelian p -subgroup of G has rank ≤ 1 .
- (iv) The p -Sylow subgroups of G are cyclic or generalized quaternion.

We close this section by stating without proof a beautiful generalization of this theorem, which was conjectured by Atiyah and Swan and proved by Quillen [1971]. Fix a prime p and let $H(G)$ denote the commutative $\mathbb{Z}_p$ -algebra $\bigoplus_{n \geq 0} H^{2n}(G, \mathbb{Z}_p)$. If the maximal rank of an elementary abelian p -subgroup of G is 1, then 9.7 implies that $H(G)$ is a finitely generated module over a polynomial subalgebra $\mathbb{Z}_p[u]$. Consequently, $H(G)$ has Krull dimension 1. (See, for example, Serre [1965] for a treatment of dimension theory of commutative rings.) Quillen's generalization is:

(9.8) Theorem. *For any finite group G , the Krull dimension of $H(G)$ is equal to the maximal rank of an elementary abelian p -subgroup of G .*

Roughly speaking, then, the complexity of the ring $H^*(G, \mathbb{Z}_p)$ is determined by the complexity of the p -subgroups of G .

EXERCISES

1. If G is non-trivial and has periodic cohomology of period d , prove that d is even. [Hint: Use anti-commutativity.]
2. Prove that G has periodic cohomology of period 2 if and only if G is cyclic. [Hint: What can you say about $H_1 G$ if $\hat{H}^*(G)$ has period 2?]
3. Compute $H^n(G, \mathbb{Z})$ and $H_n(G, \mathbb{Z})$ for all n if $\hat{H}^*(G)$ is periodic of period 4. Note, in particular, that $H_2(G) = \hat{H}^{-3}(G) = \hat{H}^1(G) = \text{Hom}(G, \mathbb{Z}) = 0$, so we obtain a new proof of the result of exercise 7a of §II.5.
4. If G has periodic cohomology, show that $\hat{H}^i(G, \mathbb{Z}) = 0$ for i odd. [Hint: It suffices to do this when G is a p -group.]
5. Give a direct proof that (iv) $\Rightarrow$ (i) in 9.3. [Hint: Use induced representations as in example 3 of 9.2.]
6. Let $G = \mathbb{Z}_m \rtimes \mathbb{Z}_n$, where m and n are relatively prime and $\mathbb{Z}_n$ acts on $\mathbb{Z}_m$ via a homomorphism $\mathbb{Z}_n \rightarrow \mathbb{Z}_m^*$ whose image has order k . Prove that the minimal period of $\hat{H}^*(G)$ is $2k$. [Hint: If $p|n$ then $\hat{H}^*(G, \mathbb{Z})_{(p)} \approx \hat{H}^*(\mathbb{Z}_n, \mathbb{Z})_{(p)}$. If $p|m$ then $\hat{H}^*(G)_{(p)} \approx \hat{H}^*(\mathbb{Z}_m)_{(p)}^{\mathbb{Z}_n}$.]
7. Let $\mathbb{F}_q$ be a field with q elements, where q is a prime power. Show that $SL_n(\mathbb{F}_q)$ does not have periodic cohomology if $n \geq 3$ or if q is not prime.

8. Let $k = \mathbb{F}_q$, where q is an odd prime power.

(a) Let M be the subgroup of $SL_2(k)$ consisting of the monomial matrices

$$\begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & -\lambda^{-1} \\ \lambda & 0 \end{pmatrix}, \quad \lambda \in k^*.$$

Show that $M \approx Q_{2(q-1)}$ if $q \neq 3$.

(b) Let $K \supset k$ be a quadratic extension (so $K = \mathbb{F}_{q^2}$). We regard K as a 2-dimensional vector space over k and denote by $GL(K)$ (resp. $SL(K)$) the group of vector space automorphisms of K (resp. the group of automorphisms of determinant 1). Thus $GL(K) \approx GL_2(k)$ and $SL(K) \approx SL_2(k)$. We have $\text{Gal}(K/k) \subset GL(K)$ and $K^* \hookrightarrow GL(K)$, where the latter embedding corresponds to the multiplication action of K^* on K . Show that $\text{Gal}(K/k)$ and K^* generate a subgroup of $GL(K)$ of order $2(q^2 - 1)$, whose intersection with $SL(K)$ is isomorphic to $Q_{2(q+1)}$. [Hint: The composite $K^* \hookrightarrow GL(K) \xrightarrow{\det} k^*$ is simply the norm map of Galois theory, which in this case is given by $\lambda \mapsto \lambda^{q+1}$. Also, if $\sigma \in \text{Gal}(K/k)$ is the non-trivial element, then $\det \sigma = -1$.]

(c) Using (a) and (b) and the fact that $|SL_2(k)| = q(q^2 - 1)$, show that $SL_2(k)$ has a quaternionic 2-Sylow subgroup and a cyclic l -Sylow subgroup for any odd prime $l \neq \text{char } k$.

9. Suppose that G has p -periodic cohomology. Let $P \subseteq G$ be a subgroup of order p , let $N(P)$ (resp. $C(P)$) be the normalizer (resp. centralizer) of P in G , and let $W = N(P)/C(P)$. Prove that $\hat{H}^*(G, M)_{(p)} \approx \hat{H}^*(N(P), M)_{(p)} \approx \hat{H}^*(C(P), M)_{(p)}^W$. [Hint: For the first isomorphism, choose a p -Sylow subgroup $H \supseteq P$ and show that every $N(P)$ -invariant in $\hat{H}^*(H)$ is G -invariant. For the second isomorphism, note that $p \nmid |W|$.]

10. For any finite group G , show that the augmentation ideal $I \subset \mathbb{Z}G$ is a cyclic G -module iff G is a cyclic group. [Hint: If I is cyclic, then G admits a periodic resolution of period 2.]

CHAPTER VII

Equivariant Homology and Spectral Sequences

1 Introduction

If a group G operates on a topological space X , then one can define *equivariant homology* and *cohomology* groups, which can be thought of heuristically as a “mixture” of $H(G)$ and $H(X)$. This equivariant theory provides a powerful tool for extracting homological information about G from the action of G on X . It is in this way, for example, that Quillen proved his theorem about the Krull dimension of $H^*(G, \mathbb{Z}_p)$ for G finite (VI.9.8).

The main purpose of this chapter is to construct the equivariant homology theory and give its basic properties, including two spectral sequences. This theory will play a crucial role in the remaining chapters of this book. For simplicity, we will confine ourselves to the case where X is a G -complex in the sense of §I.4. See Grothendieck [1957] or Quillen [1971] for a more general point of view.

We begin with a brief treatment of the theory of spectral sequences.

2 The Spectral Sequence of a Filtered Complex

If C is a chain complex and C' is a subcomplex, then there is a long exact sequence which gives information about $H_*(C)$ in terms of $H_*(C')$ and $H_*(C/C')$. Now suppose we are given, instead of a single subcomplex C' , a sequence of subcomplexes $\{F_p C\}_{p \in \mathbb{Z}}$, with $F_{p-1} C \subseteq F_p C$. It is reasonable, then, to try to get information about $H_*(C)$ in terms of the groups $H_*(F_p C/F_{p-1} C)$. In this section, we will describe a method for doing this.

What one obtains, roughly speaking, is a sequence of successive approximations E^r ($r \geq 0$) to $H_*(C)$, such that E^1 consists of the groups $H_*(F_p C / F_{p-1} C)$.

We will omit most of the proofs in this section. The reader can either supply the missing proofs (which are routine) or consult any text which treats spectral sequences, e.g., Spanier [1966] or MacLane [1963].

Let R be an arbitrary ring. (In our applications, we will usually take $R = \mathbb{Z}$.) By an *increasing filtration* on an R -module M we mean a family of submodules $F_p M$ ($p \in \mathbb{Z}$) such that $F_p M \subseteq F_{p+1} M$. The filtration is said to be *finite* if $F_p M = 0$ for p sufficiently small and $F_p M = M$ for p sufficiently large. Given a filtration on M , the *associated graded module* $\text{Gr } M$ is defined by $\text{Gr}_p M = F_p M / F_{p-1} M$. One thinks of M , then, as being built up from the “pieces” $\text{Gr}_p M$. The following elementary lemma shows that, in some sense, we do not lose too much information by passing from M to $\text{Gr } M$:

(2.1) Lemma. *Let $f: M \rightarrow M'$ be a filtration-preserving map, where M and M' are modules with finite filtrations. If $\text{Gr } f: \text{Gr } M \rightarrow \text{Gr } M'$ is an isomorphism, then f is an isomorphism.*

Another elementary (but important) observation is that if we have a notion of “rank” for R -modules (e.g., if R is a field or $R = \mathbb{Z}$), such that $\text{rk } M = \text{rk } M' + \text{rk } M''$ for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$, then the rank of a finitely-filtered module can be computed from the associated graded module:

$$(2.2) \quad \text{rk } M = \sum_{p \in \mathbb{Z}} \text{rk } \text{Gr}_p M.$$

If the filtered module M is itself graded (and each $F_p M$ is a graded submodule), then we have for each $n \in \mathbb{Z}$ a filtration $\{F_p M_n\}$ on M_n , and hence there is an obvious way of associating a *bigraded* module to M . The usual notational convention in this case is to set $\text{Gr}_{pq} M = F_p M_{p+q} / F_{p-1} M_{p+q}$. An element of $\text{Gr}_{pq} M$ is said to have *filtration degree* p , *complementary degree* q , and *total degree* $p + q$. To simplify the notation, we will sometimes suppress the second subscript and simply write $\text{Gr}_p M = F_p M / F_{p-1} M$.

Now let $C = (C_n)_{n \in \mathbb{Z}}$ be a filtered chain complex (with each $F_p C$ a subcomplex). For simplicity, we will always assume that the filtration is *dimension-wise finite*, i.e., that $\{F_p C_n\}_{p \in \mathbb{Z}}$ is a finite filtration of C_n for each n . There is an induced filtration on the homology $H(C)$, defined by $F_p H(C) = \text{Im}\{H(F_p C) \rightarrow H(C)\}$. We can identify $F_p H(C)$ with $(F_p C \cap Z) / (F_p C \cap B)$, where Z (resp. B) is the module of cycles (resp. boundaries) of C . The associated bigraded module $\text{Gr } H(C)$ is given by

$$(2.3) \quad \text{Gr}_p H(C) = (F_p C \cap Z) / ((F_p C \cap B) + (F_{p-1} C \cap Z)).$$

We now describe the *spectral sequence* associated to the filtered complex

C. This is a sequence $\{E^r\}_{r \geq 0}$ of “successive approximations” to $\text{Gr } H(C)$. Let $Z_p^r = F_p C \cap \partial^{-1} F_{p-r} C$. (More precisely,

$$Z_{pq}^r = F_p C_{p+q} \cap \partial^{-1} F_{p-r} C_{p+q-1}.)$$

Let $Z_p^\infty = F_p C \cap Z$. We have $F_p C = Z_p^0 \supseteq Z_p^1 \supseteq \cdots \supseteq Z_p^\infty$. Since the filtration $\{F_p C\}$ is dimension-wise finite, this sequence of inclusions stabilizes dimension-wise to a sequence of equalities, i.e., for fixed (p, q) we have $Z_{pq}^r = Z_{pq}^{r+1} = \cdots = Z_{pq}^\infty$ for r sufficiently large. Let $B_p^r = F_p C \cap \partial F_{p+r-1} C = \partial Z_{p+r-1}^{r-1}$, and let $B_p^\infty = F_p C \cap B$. Then $B_p^0 \subseteq B_p^1 \subseteq \cdots \subseteq B_p^\infty$, and again this stabilizes dimension-wise to a sequence of equalities. We now have

$$B_p^0 \subseteq B_p^1 \subseteq \cdots \subseteq B_p^\infty \subseteq Z_p^\infty \subseteq \cdots \subseteq Z_p^1 \subseteq Z_p^0 = F_p C,$$

and we set

$$E_p^r = Z_p^r / (B_p^r + Z_p^{r-1}) = Z_p^r / (B_p^r + (F_{p-1} C \cap Z_p^r))$$

and

$$E_p^\infty = Z_p^\infty / (B_p^\infty + Z_p^\infty) = \text{Gr}_p H(C)$$

(cf. 2.3). For fixed (p, q) , we have

$$E_{pq}^r = E_{pq}^{r+1} = \cdots = E_{pq}^\infty$$

for r sufficiently large, so the sequence $\{E^r\}$ “converges” to $\text{Gr } H(C)$ as $r \rightarrow \infty$. (One often suppresses the “Gr” here and simply says that the spectral sequence *converges to* $H(C)$ or that $H(C)$ is the *abutment* of the spectral sequence.)

The modules E^r are particularly easy to describe for $r = 0$ and 1:

$$(2.4) \quad E_p^0 = F_p C / F_{p-1} C = \text{Gr}_p C$$

$$(2.5) \quad E_p^1 = (F_p C \cap \partial^{-1} F_{p-1} C) / (\partial F_p C + F_{p-1} C) \approx H(F_p C / F_{p-1} C).$$

(More precisely, $E_{pq}^1 \approx H_{p+q}(F_p C / F_{p-1} C)$.) Thus E^1 is the homology of E^0 , relative to the differential induced on E^0 by ∂ . More generally, one can show that ∂ induces a differential d^r on E^r of bidegree $(-r, r-1)$ (i.e., $d^r: E_{pq}^r \rightarrow E_{p-r, q+r-1}^r$), and that $E^{r+1} \approx H(E^r)$. An important consequence of this is:

(2.6) Proposition. *Let $\tau: C \rightarrow C'$ be a filtration-preserving chain map, where C and C' have dimension-wise finite filtrations. If the induced map $E^r(\tau): E^r(C) \rightarrow E^r(C')$ of spectral sequences is an isomorphism for some r , then $H(\tau): H(C) \rightarrow H(C')$ is an isomorphism.*

PROOF. If $E^r(\tau)$ is an isomorphism then so is $E^s(\tau)$ for $s > r$, since $E^s = H(E^{s-1})$. Hence $E^\infty(\tau) = \text{Gr } H(\tau)$ is an isomorphism, and the proposition now follows from 2.1. $\square$

Another important fact is that spectral sequences can be used to compute Euler characteristics. Suppose, for example, that $R = \mathbb{Z}$ or R is a field. For

any graded R -module $A = (A_n)$ (resp. bigraded module $A = (A_{pq})$), let $\chi(A) = \sum_n (-1)^n \text{rk } A_n$ (resp. $\chi(A) = \sum_{p,q} (-1)^{p+q} \text{rk } A_{pq}$), provided all the ranks are finite and almost all of them are zero. In case A has a differential, it is well-known that $\chi(A) = \chi(H(A))$ (cf. Spanier [1966], 4.3.14). In particular, if C is a filtered complex as above and if $\chi(E^r)$ is defined for some r , then it follows that

$$(2.7) \quad \chi(E^r) = \chi(E^{r+1}) = \cdots = \chi(E^\infty) = \chi(H(C)),$$

the last equality being a consequence of 2.2.

Finally, we briefly describe the notational conventions in case C is indexed as a *cochain complex* $(C^n)_{n \in \mathbb{Z}}$, with differential of degree $+1$. In this case the filtration is denoted $\{F^p C\}$ and is assumed to be *decreasing* (i.e., $F^p C \supseteq F^{p+1} C$). The terms of the resulting *cohomology spectral sequence* are denoted E_r^{pq} , and one has $E_\infty^{pq} = \text{Gr}^{pq} H(C) = F^p H^{p+q}(C) / F^{p+1} H^{p+q}(C)$. The differential d_r is of bidegree $(r, -r+1)$, so that $d_r: E_r^{pq} \rightarrow E_r^{p+r, q-r+1}$. If we think of E_r^{pq} as sitting on the lattice point (p, q) of the plane, then d_r can be visualized as an arrow pointing to the right and downward for $r \geq 2$. By contrast, the differentials d^r in a homology spectral sequence point to the left and upward.

The canonical example of a filtered cochain complex is $\mathcal{H}om(C, M)$, where C is a filtered chain complex. One sets

$$F^p \mathcal{H}om(C, M) = \mathcal{H}om(C/F_{p-1} C, M).$$

EXERCISE

Let $0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$ be a short exact sequence of chain complexes. Let $\{F_p C\}$ be the filtration such that $F_0 C = 0$, $F_1 C = C'$, and $F_2 C = C$. Describe the modules E_{pq}^1 and deduce from the spectral sequence the familiar long exact homology sequence. [Thus the spectral sequence of a filtered complex can be regarded as a generalization of the long exact sequence associated to a chain complex and a subcomplex, as we implied in the introductory paragraph of this section.]

3 Double Complexes

By a *double complex* we mean a bigraded module $C = (C_{pq})_{p,q \in \mathbb{Z}}$ with a “horizontal” differential ∂' of bidegree $(-1, 0)$ and a “vertical” differential ∂'' of bidegree $(0, -1)$, such that $\partial' \partial'' = \partial'' \partial'$:

$$\begin{array}{ccc} C_{p-1,q} & \xleftarrow{\partial'} & C_{pq} \\ \partial'' \downarrow & & \downarrow \partial'' \\ C_{p-1,q-1} & \xleftarrow{\partial'} & C_{p,q-1} \end{array}$$

Note that a double complex can be regarded, in two different ways, as a "chain complex in the category of chain complexes." Thus for each q we have a horizontal chain complex $C_{*,q}$ with differential ∂' , and we are given chain maps $\partial'': C_{*,q} \rightarrow C_{*,q-1}$ such that $\partial''\partial'' = 0$. Similarly, for each p we have a vertical chain complex $C_{p,*}$ with differential ∂'' , and we are given chain maps $\partial': C_{p,*} \rightarrow C_{p-1,*}$ with $\partial'\partial' = 0$.

A double complex C gives rise to an ordinary chain complex TC , called the *total complex*, as follows: $(TC)_n = \bigoplus_{p+q=n} C_{pq}$, with differential ∂ given by $\partial|C_{pq} = \partial' + (-1)^p\partial''$. The tensor product of two chain complexes C' and C'' provides a familiar example of this construction. Indeed, we have a double complex C with $C_{pq} = C'_p \otimes C''_q$, and TC is simply the usual tensor product $C' \otimes C''$ of chain complexes.

We now filter TC by setting $F_p(TC)_n = \bigoplus_{i \leq p} C_{i,n-i}$. This is a dimension-wise finite filtration provided C has only finitely many non-zero modules C_{pq} in any given total degree $p+q$. (This holds automatically, for example, if C is a *first quadrant* double complex, i.e., if $C_{pq} = 0$ when $p < 0$ or $q < 0$.) Thus we have a spectral sequence $\{E^r\}$ converging to $H_*(TC)$. It is immediate from the definitions that $E_{pq}^0 = C_{pq}$ and that $d^0 = \pm \partial''$. Consequently, E^1 is the vertical homology of C , i.e., $E_{pq}^1 = H_q(C_{p,*})$. The differential $d^1: E_{pq}^1 \rightarrow E_{p-1,q}^1$ is easily seen to be the map induced by the chain map $\partial': C_{p,*} \rightarrow C_{p-1,*}$; for an element of E_{pq}^1 is represented by an element $c \in C_{pq}$ such that $\partial''c = 0$, and for such a c one has $\partial c = \partial'c$. Thus E^2 can be described as the horizontal homology of the vertical homology of C .

One could equally well filter TC by $F_p(TC)_n = \bigoplus_{j \leq p} C_{n-j,j}$. We obtain, then, a second spectral sequence converging to $H(TC)$, this time with $E_{pq}^0 = C_{qp}$, $E_{pq}^1 = H_q(C_{*,p})$, and $d^1: E_{pq}^1 \rightarrow E_{p-1,q}^1$ equal (up to sign) to the map induced by $\partial'': C_{*,p} \rightarrow C_{*,p-1}$.

[Warning: Even though the two spectral sequences have the same abutment $H_*(TC)$, they do not in general have the same E^∞ -term; for we have two different filtrations on $H(TC)$ and hence two different E^∞ -terms $\text{Gr } H(TC)$.]

A similar discussion applies to double cochain complexes $C = C^{pq}$. One has a total cochain complex TC with two decreasing filtrations, and hence two spectral sequences of cohomological type converging to $H^*(TC)$ (provided that C has only finitely many non-zero modules in any given total degree). Details are left to the reader.

EXERCISE

Let C be a first-quadrant double complex such that one of the associated spectral sequences (the first one, say) has $E_{pq}^1 = 0$ for $q \neq 0$. Let D be the chain complex $E_{*,0}^1$ with differential d^1 .

(a) Show that there is an isomorphism $\varphi: H_*(TC) \xrightarrow{\sim} H_*(D)$.

(b) Make this result more precise by showing that there is a weak equivalence $\tau: TC \rightarrow D$. [Hint: Take τ to be the canonical surjection which comes from the fact that D is the vertical 0-dimensional homology of C . Show that $\tau_* = \varphi$ by directly examining the definitions in the construction of the spectral sequence. Alternatively, you can avoid getting your hands dirty by observing that τ can be viewed as a map of double complexes $C \rightarrow D$ (where D is regarded as a double complex concentrated on the line $q = 0$); the induced map of spectral sequences is an isomorphism at the E^1 -level, so τ induces an isomorphism $H_*(TC) \rightarrow H_*(TD) = H_*(D)$.]

4 Example: The Homology of a Union

Let X be a CW -complex which is the union of a family of non-empty subcomplexes X_α , where α ranges over some totally ordered index set J . In case $J = \{1, 2\}$, one has a short exact sequence of chain complexes

$$0 \rightarrow C(X_1 \cap X_2) \rightarrow C(X_1) \oplus C(X_2) \rightarrow C(X) \rightarrow 0$$

(where $C(\quad)$ denotes the *cellular* chain complex), from which one obtains the familiar Mayer–Vietoris sequence. In the general case, we will show that the exact sequence above is replaced by an exact sequence

$$\cdots \rightarrow \bigoplus_{\alpha < \beta < \gamma} C(X_\alpha \cap X_\beta \cap X_\gamma) \rightarrow \bigoplus_{\alpha < \beta} C(X_\alpha \cap X_\beta) \rightarrow \bigoplus_{\alpha} C(X_\alpha) \rightarrow C(X) \rightarrow 0$$

and that the Mayer–Vietoris sequence is replaced by a spectral sequence. The details are as follows.

Let K be the abstract simplicial complex whose vertex set is J and whose simplices are the non-empty finite subsets σ of J such that the intersection $X_\sigma = \bigcap_{\alpha \in \sigma} X_\alpha$ is non-empty; K is called the *nerve* of the covering $\{X_\alpha\}$. For $p \geq 0$ let C_p be the chain complex $\bigoplus_{\sigma \in K^{(p)}} C(X_\sigma)$, where $K^{(p)}$ is the set of p -simplices of K . If σ has vertices $\alpha_0 < \cdots < \alpha_p$, we denote by $\partial_i \sigma$ ($0 \leq i \leq p$) the $(p-1)$ -simplex $\{\alpha_0, \dots, \hat{\alpha}_i, \dots, \alpha_p\}$. The inclusions $C(X_\sigma) \hookrightarrow C(X_{\partial_i \sigma})$ induce a chain map $\partial_i: C_p \rightarrow C_{p-1}$ for $p \geq 1$, and we set $\partial = \sum_{i=0}^p (-1)^i \partial_i$. Similarly, the inclusions $C(X_\alpha) \hookrightarrow C(X)$ induce a chain map $\varepsilon: C_0 \rightarrow C(X)$. We have, then, an augmented chain complex

$$(4.1) \quad \cdots \rightarrow C_p \xrightarrow{\partial} C_{p-1} \rightarrow \cdots \rightarrow C_0 \xrightarrow{\varepsilon} C(X) \rightarrow 0$$

in the category of chain complexes, and hence a double complex C with C_{pq} equal to the group of q -chains of C_p , i.e., $C_{pq} = \bigoplus_{\sigma \in K^{(p)}} C_q(X_\sigma)$.

I claim now that 4.1 is exact. To see this, we give an alternative description of C_{pq} . For any cell e of X , let K_e be the subcomplex of K consisting of the simplices σ such that $e \subseteq X_\sigma$. Then C_{pq} has one basis element for every pair (σ, e) such that e is a q -cell of X and σ is a p -simplex of K_e . In other words, $C_{pq} \approx \bigoplus_{e \in X^{(q)}} C_p(K_e)$, where $X^{(q)}$ is the set of q -cells of X . Moreover, an examination of the definitions of the maps ∂ and ε in 4.1 shows that $\partial: C_{pq} \rightarrow$

$C_{p-1,q}$ is equal to $\bigoplus_{e \in X^{(q)}} \{\partial: C_p(K_e) \rightarrow C_{p-1}(K_e)\}$, and similarly for ε . Thus the q -dimensional part of 4.1,

$$\cdots \rightarrow C_{pq} \rightarrow \cdots \rightarrow C_{0q} \rightarrow C_q(X) \rightarrow 0,$$

is isomorphic to $\bigoplus_{e \in X^{(q)}} \tilde{C}(K_e)$, where $\tilde{C}(K_e)$ is the augmented simplicial chain complex of K_e . But K_e is acyclic. Indeed, its simplices are all of the finite subsets of the non-empty set $J_e = \{\alpha \in J: e \subseteq X_\alpha\}$, so K_e is the “simplex” spanned by the (possibly infinite) set J_e . Thus $\bigoplus_e \tilde{C}(K_e)$ is acyclic, which proves the claim.

Consider, now, the two spectral sequences of the double complex (C_{pq}) . In view of the exactness of 4.1, the second spectral sequence has

$$E_{pq}^1 = H_q(C_{*,p}) = \begin{cases} 0 & \text{if } q \neq 0 \\ C_p(X) & \text{if } q = 0. \end{cases}$$

Taking the homology with respect to p , we obtain

$$E_{pq}^2 = \begin{cases} 0 & \text{if } q \neq 0 \\ H_p(X) & \text{if } q = 0. \end{cases}$$

The spectral sequence therefore “collapses” to yield:

$$(4.2) \quad H_*(TC) \approx H_*(X).$$

The first spectral sequence, on the other hand, has

$$(4.3) \quad E_{pq}^1 = H_q(C_p) = \bigoplus_{\sigma \in K^{(p)}} H_q(X_\sigma).$$

Moreover, in view of the isomorphism 4.2, we can regard $H_*(X)$ as the abutment of the spectral sequence, i.e., $E^\infty \approx \text{Gr } H_*(X)$ relative to some filtration on $H_*(X)$. This first spectral sequence, then, is our desired generalization of the Mayer–Vietoris sequence. It approximates H_*X in terms of the homology of the X_σ and their intersections.

To describe the E^2 -term, we need the notion of “coefficient system” on a simplicial complex K . By this we will mean a family $\mathcal{A} = \{A_\sigma\}$ of abelian groups, where σ ranges over the simplices of K , together with a map $f_{\sigma\tau}: A_\sigma \rightarrow A_\tau$ whenever τ is a face of σ (written $\tau \subset \sigma$), such that $f_{\tau\rho} f_{\sigma\tau} = f_{\sigma\rho}$ if $\rho \subset \tau \subset \sigma$. There is an obvious way to construct a chain complex $C(K, \mathcal{A})$, with $C_p(K, \mathcal{A}) = \bigoplus_{\sigma \in K^{(p)}} A_\sigma$, and one therefore has homology groups $H_*(K, \mathcal{A})$. (See Godement [1958], I.3.3, for a detailed treatment of the cohomological version of this.)

Returning now to the situation where K is the nerve of the covering $\{X_\alpha\}$, we have for each $q \geq 0$ a coefficient system $\mathcal{H}_q = \{H_q(X_\sigma)\}$ on K , where $f_{\sigma\tau}: H_q(X_\sigma) \rightarrow H_q(X_\tau)$ is induced by the inclusion $X_\sigma \hookrightarrow X_\tau$. It is immediate from the definition that the E^1 -term in 4.3 is equal to $C_p(K, \mathcal{H}_q)$, hence $E_{pq}^2 = H_p(K, \mathcal{H}_q)$. We summarize this discussion by writing

$$E_{pq}^2 = H_p(K, \mathcal{H}_q) \Rightarrow H_{p+q}(X),$$

where the symbol “ $\Rightarrow$ ” indicates that the spectral sequence converges to what appears on the right-hand side.

In case each X_σ is acyclic, we have $\mathcal{H}_0 = \mathbb{Z}$ and $\mathcal{H}_q = 0$ for $q \neq 0$. The spectral sequence therefore collapses at E^2 to yield the following result, which seems to be essentially due to Leray:

(4.4) Theorem. *Suppose X is the union of subcomplexes X_α such that every non-empty intersection $X_{\alpha_0} \cap \cdots \cap X_{\alpha_p}$ ($p \geq 0$) is acyclic. Then $H_*(X) \approx H_*(K)$, where K is the nerve of the cover.*

EXERCISES

1. Make 4.4 more precise by showing that, under the given hypotheses, there is a chain complex T which admits weak equivalences $T \rightarrow C(X)$ and $T \rightarrow C(K)$. [Take $T = TC$, where C is the double complex used in this section, and use part (b) of the exercise of §3.]
- *2. Make 4.4 still more precise by showing that there is a CW -complex Y which maps to both K and X by maps inducing homology isomorphisms. [Hint: Let Y be the union of the cells $\sigma \times e$ in $K \times X$ such that $e \subseteq X_\sigma$ (or, equivalently, such that σ is in K_e); note that $C(Y)$ can be identified with the complex T of exercise 1.]

Remark. The map $Y \rightarrow X$ is in fact a homotopy equivalence. This can be seen, for example, by an inductive argument over the skeleta of X , using the fact that each complex K_e is contractible. Consequently, one obtains a canonical homotopy class of maps $X \rightarrow K$ inducing a homology isomorphism. In case each X_σ is contractible (and not just acyclic), one can similarly show that the map $Y \rightarrow K$ is a homotopy equivalence, so that X and K are homotopy equivalent. This sort of result was first proved by Weil [1952], §5; see also Borel–Serre [1974], §8, and Quillen [1978], 1.6–1.8.

5 Homology of a Group with Coefficients in a Chain Complex

Recall that $H_*(G, M)$ is defined to be $H_*(F \otimes_G M)$, where F is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. It is useful to generalize this by allowing a non-negative chain complex $C = (C_n)_{n \geq 0}$ of coefficients. Thus we set

$$H_*(G, C) = H_*(F \otimes_G C);$$

as usual, this is well-defined up to canonical isomorphism. If C consists of a single module M concentrated in dimension 0, then $H_*(G, C)$ reduces to $H_*(G, M)$.

Since $F \otimes_G C$ is the total complex of the double complex of abelian groups $(F_p \otimes_G C_q)$, we have by §3 two spectral sequences converging to $H_*(G, C)$. The first of these has $E_{pq}^1 = H_q(F_p \otimes_G C_*) = F_p \otimes_G H_q(C)$ since $F_p \otimes_G -$ is an exact functor. Taking now the homology with respect to p , we obtain $E_{pq}^2 = H_p(G, H_q C)$. Thus the spectral sequence has the form

$$(5.1) \quad E_{pq}^2 = H_p(G, H_q C) \Rightarrow H_{p+q}(G, C).$$

An important consequence of this is that $H_*(G, C)$ is an invariant of the "weak homotopy type" of C :

(5.2) Proposition. *If $\tau: C \rightarrow C'$ is a weak equivalence of G -chain complexes, then τ induces an isomorphism $H_*(G, C) \xrightarrow{\sim} H_*(G, C')$.*

PROOF. τ induces a map of spectral sequences which is an isomorphism at the E^2 -level, hence τ induces an isomorphism on the abutments by 2.6. $\square$

[This result also follows, of course, from I.8.6, but it still provides a nice illustration of spectral sequence techniques.]

The second spectral sequence has $E_{pq}^1 = H_q(F_* \otimes_G C_p) = H_q(G, C_p)$. Thus we have:

$$(5.3) \quad E_{pq}^1 = H_q(G, C_p) \Rightarrow H_{p+q}(G, C).$$

The group E_{pq}^2 can therefore be described as the p -th homology group of the complex obtained from C by applying the functor $H_q(G, -)$ dimension-wise. Both spectral sequences, then, can be thought of as giving approximations to $H_*(G, C)$ in terms of *ordinary* homology groups $H_*(G, M)$.

To get a better feeling for $H_*(G, C)$, let's look at some special cases. Suppose, first, that G acts trivially on C . Then $F \otimes_G C \approx F_G \otimes C$, so there is a *Künneth formula*

$$0 \rightarrow \bigoplus_{p+q=n} H_p G \otimes H_q C \rightarrow H_n(G, C) \rightarrow \bigoplus_{p+q=n-1} \text{Tor}(H_p G, H_q C) \rightarrow 0$$

expressing $H_*(G, C)$ in terms of $H_* G$ and $H_* C$. Similarly, if k is a field and C is a complex of k -vector spaces (with trivial G -action), then

$$(5.4) \quad H_*(G, C) \approx H_*(G, k) \otimes_k H_* C.$$

At the other extreme, suppose that each C_p is a *free* $\mathbb{Z}G$ -module, or, more generally, an H_* -acyclic G -module (e.g., projective or induced). The E^1 -term in 5.3 is then concentrated on the line $q = 0$, and $E_{p,0}^1 = (C_p)_G$. The spectral sequence therefore collapses at E^2 to yield:

$$(5.5) \quad H_*(G, C) \approx H_*(C_G).$$

The abutment of the spectral sequence (5.1) can therefore be identified with $H_*(C_G)$; this proves:

(5.6) Proposition. *Let C be a non-negative chain complex of G -modules such that each C_n is H_* -acyclic. Then there is a spectral sequence of the form*

$$E_{pq}^2 = H_p(G, H_q C) \Rightarrow H_{p+q}(C_G).$$

Remark. This spectral sequence is a special case of the “universal coefficient spectral sequence,” cf. Godement [1958], I.5.5.1.

We can also define *cohomology groups* $H^*(G, C)$, where $C = (C^n)_{n \geq 0}$ is a non-negative cochain complex; namely, we set

$$H^*(G, C) = H^*(\mathcal{H}om_G(F, C)),$$

where F is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$. Now $\mathcal{H}om_G(F, C)$ is in fact the total complex associated to a double complex with $C^{pq} = \text{Hom}_G(F_q, C^p)$. Indeed, we have $\mathcal{H}om_G(F, C)^n = \mathcal{H}om_G(F, C)_{-n} = \prod_{p+q=n} \text{Hom}_G(F_q, C_{-p}) = \bigoplus_{p+q=n} \text{Hom}_G(F_q, C^p)$. And if we take δ' to be the coboundary operator in $\mathcal{H}om_G(F_q, C)$ [where q is fixed and the module F_q is regarded as a chain complex concentrated in dimension 0] and δ'' to be the coboundary operator in $\mathcal{H}om_G(F, C^p)$, then the resulting total coboundary operator is the same as that in $\mathcal{H}om_G(F, C)$. One can now easily deduce cohomology analogues of 5.1–5.6.

Similarly, if G is finite, one can define *Tate cohomology groups* $\hat{H}^*(G, C)$ by using a *complete* resolution F . We assume, here, that $C^n = 0$ for $n \gg 0$, so that the double complex $\text{Hom}_G(F_p, C^q)$ has only finitely many non-zero groups in each total degree.

Finally, we remark that all of the formal properties of homology and cohomology given in Chapters III and V extend without difficulty to homology and cohomology with coefficients in a (co-) chain complex. In particular, one has long exact sequences, Shapiro’s lemma, restriction and corestriction maps, cup products, etc. Moreover, the cohomology spectral sequences analogous to 5.1 and 5.3 are compatible with the cup products. More precisely, suppose C and D are G -cochain complexes, and consider the cochain cup product

$$\mathcal{H}om_G(F, C) \otimes \mathcal{H}om_G(F, D) \rightarrow \mathcal{H}om_G(F, C \otimes D)$$

defined via some diagonal approximation $F \rightarrow F \otimes F$. One checks easily that this cup product is compatible with the filtrations defining the two spectral sequences, i.e., that it induces a product

$$F^p \mathcal{H}om_G(F, C) \times F^{p'} \mathcal{H}om_G(F, D) \rightarrow F^{p+p'} \mathcal{H}om_G(F, C \otimes D).$$

It follows that there is an induced cup product

$$E_r^{pq}(C) \otimes E_r^{p'q'}(D) \rightarrow E_r^{p+p', q+q'}(C \otimes D),$$

where $\{E_r(\quad)\}$ denotes the cohomology version of either 5.1 or 5.3. We will see applications of this product structure in Chapter X.

6 Example: The Hochschild–Serre Spectral Sequence

Let $1 \rightarrow H \rightarrow G \rightarrow Q \rightarrow 1$ be a group extension. If F is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$ and M is a G -module, then $F \otimes_G M = (F \otimes M)_G$ can be computed in two steps (cf. §II.2, exercise 3), by first dividing out by the action of H on $F \otimes M$ and then dividing out by the action of Q :

$$F \otimes_G M = ((F \otimes M)_H)_Q = (F \otimes_H M)_Q.$$

Thus

$$(6.1) \quad H_*(G, M) = H_*(C_Q),$$

where $C = F \otimes_H M$. Note also that we have a Q -module isomorphism

$$(6.2) \quad H_*(H, M) \approx H_*(C),$$

where the Q -action on $H_*(H, M)$ is that of III.8.2. Finally, I claim that the Q -modules $C_p = (F_p \otimes M)_H$ are H_* -acyclic. In fact, it suffices to show that $(\mathbb{Z}G \otimes M)_H$ is H_* -acyclic; and for this one need only observe (by using III.5.7, for example) that $(\mathbb{Z}G \otimes M)_H$ is an induced Q -module $\mathbb{Z}Q \otimes A$.

We are now in a position to apply 5.6 to the Q -complex C . In view of 6.1 and 6.2 we obtain:

(6.3) Theorem (Hochschild–Serre [1953]). *For any group extension $1 \rightarrow H \rightarrow G \rightarrow Q \rightarrow 1$ and any G -module M , there is a spectral sequence of the form*

$$E_{pq}^2 = H_p(Q, H_q(H, M)) \Rightarrow H_{p+q}(G, M).$$

(There is, of course, an analogous spectral sequence in cohomology, derived from the observation that $\mathcal{H}om_G(F, M) = (\mathcal{H}om_H(F, M))^Q$.)

An important consequence of 6.3 is the following 5-term exact sequence of low-dimensional homology groups, generalizing the 5-term exact sequence of exercise 6 of §II.5:

(6.4) Corollary. *Under the hypotheses of 6.3 there is an exact sequence*

$$H_2(G, M) \rightarrow H_2(Q, M_H) \rightarrow H_1(H, M)_Q \rightarrow H_1(G, M) \rightarrow H_1(Q, M_H) \rightarrow 0.$$

PROOF. Since $E^\infty = \text{Gr } H(G, M)$, we have a short exact sequence

$$0 \rightarrow E_{0,1}^\infty \rightarrow H_1(G, M) \rightarrow E_{1,0}^\infty \rightarrow 0.$$

Now there are no non-zero differentials involving $E_{1,0}^r$ ($r \geq 2$), so

$$E_{1,0}^\infty = E_{1,0}^2.$$

And the only possible non-zero differential involving $E'_{0,1}$ or $E'_{2,0}$ is

$$d^2: E'_{2,0} \rightarrow E'_{0,1},$$

so there is an exact sequence

$$0 \rightarrow E'_{2,0} \rightarrow E'_{2,0} \xrightarrow{d^2} E'_{0,1} \rightarrow E'_{0,1} \rightarrow 0.$$

The short-exact sequence above now yields an exact sequence

$$0 \rightarrow E'_{2,0} \rightarrow E'_{2,0} \rightarrow E'_{0,1} \rightarrow H_1(G, M) \rightarrow E'_{1,0} \rightarrow 0.$$

Since $E'_{pq} = H_p(Q, H_q(H, M))$ and $E'_{2,0}$ is a quotient of $H_2(G, M)$, this yields the desired 5-term exact sequence. $\square$

7 Equivariant Homology

We now specialize the theory of §5 to the case where the chain complex C is the cellular chain complex $C(X)$ of a G -complex X (cf. §I.4). The resulting homology groups $H_*(G, C(X))$ are denoted $H_*^G(X)$ and called the *equivariant homology groups* of (G, X) . More generally, if M is a G -module, then there is a diagonal G -action on $C(X, M) = C(X) \otimes M$, and we set

$$H_*^G(X, M) = H_*(G, C(X, M)).$$

Similarly, the *equivariant cohomology groups* are defined by

$$H_*^G(X, M) = H^*(G, C^*(X, M)).$$

In case X is finite dimensional and G is finite, we can also define *equivariant Tate cohomology groups* by

$$\hat{H}_*^G(X, M) = \hat{H}^*(G, C^*(X, M)).$$

Finally, if $Y \subseteq X$ is a G -invariant subcomplex, then there are relative equivariant homology and cohomology groups, defined by means of the relative complex $C(X, Y) = C(X)/C(Y)$.

For simplicity, we will confine ourselves in this section to a discussion of the (absolute) homology groups $H_*^G(X, M)$. There are, of course, relative versions, as well as cohomology analogues, of all the results.

Note first that $H_*^G(\text{pt.}, M) = H_*(G, M)$. Since any G -complex X admits a (unique) G -map to a point, we deduce that there is a *canonical map*

$$(7.1) \quad H_*^G(X, M) \rightarrow H_*(G, M).$$

Next we record the spectral sequence 5.1 in the present context:

$$(7.2) \quad E_{pq}^2 = H_p(G, H_q(X, M)) \Rightarrow H_{p+q}^G(X, M).$$

(Note that the E^2 -term here involves the *diagonal action* of G on $H_*(X, M)$, induced by the action of G on X and M .)

As a consequence of 7.2, we have (cf. 5.2):

(7.3) Proposition. *If $f: X \rightarrow Y$ is a cellular map of G -complexes such that $f_*: H_* X \rightarrow H_* Y$ is an isomorphism, then f induces an isomorphism $H_*^G(X, M) \cong H_*^G(Y, M)$ for any G -module M . In particular, if X is acyclic then the canonical map 7.1 is an isomorphism $H_*^G(X, M) \cong H_*(G, M)$.*

To analyze the second spectral sequence (5.3), we decompose $C_p(X) \otimes M$ as in III.5.5b. For each p -cell σ of X we have a G_σ -module $\mathbb{Z}_\sigma$ which is additively isomorphic to $\mathbb{Z}$, on which G_σ operates by the “orientation character” $\chi_\sigma: G_\sigma \rightarrow \{\pm 1\}$. Let

$$M_\sigma = \mathbb{Z}_\sigma \otimes M;$$

thus M_σ is a G_σ -module which is additively isomorphic to M , with the G_σ -action “twisted” by χ_σ . Let X_p be the set of p -cells of X and let Σ_p be a set of representatives for X_p/G . We then have an additive decomposition

$$(7.4) \quad C_p(X, M) = C_p(X) \otimes M = \bigoplus_{\sigma \in X_p} M_\sigma,$$

from which we obtain a G -module decomposition

$$(7.5) \quad C_p(X, M) \approx \bigoplus_{\sigma \in \Sigma_p} \text{Ind}_{G_\sigma}^G M_\sigma.$$

Shapiro’s lemma now yields

$$(7.6) \quad H_q(G, C_p(X, M)) \approx \bigoplus_{\sigma \in \Sigma_p} H_q(G_\sigma, M_\sigma),$$

so that 5.3 takes the form:

$$(7.7) \quad E_{pq}^1 = \bigoplus_{\sigma \in \Sigma_p} H_q(G_\sigma, M_\sigma) \Rightarrow H_{p+q}^G(X, M).$$

Suppose, for example, that the G -action is free, so that each $G_\sigma = \{1\}$, and assume for simplicity that $M = \mathbb{Z}$. The spectral sequence then collapses at E^2 to yield (cf. 5.5):

$$(7.8) \quad H_*^G(X) \approx H_*(C(X)_G) = H_*(X/G).$$

Combining 7.8 and 7.2, we conclude (cf. 5.6):

(7.9) Theorem. *If X is a free G -complex, then there is a spectral sequence of the form*

$$E_{pq}^2 = H_p(G, H_q X) \Rightarrow H_{p+q}(X/G).$$

Remark. This spectral sequence is called the *Cartan–Leray spectral sequence* associated to the regular covering map $X \rightarrow X/G$. Theorem 7.9 remains true if we introduce an arbitrary G -module M of coefficients, but the resulting homology groups $H_*(X/G, M)$ have to be interpreted as homology groups with local coefficients if G acts non-trivially on M .

Consider now the situation where X is acyclic (but the action of G is not necessarily free), so that

$$H_*^G(X, M) \approx H_*(G, M)$$

by 7.3. The spectral sequence 7.7 can then be rewritten:

$$(7.10) \quad E_{pq}^1 = \bigoplus_{\sigma \in \Sigma_p} H_q(G_\sigma, M_\sigma) \Rightarrow H_{p+q}(G, M).$$

This is an important computational tool in the homology theory of groups. To use it, of course, one must find an interesting acyclic space X on which G acts. If G is an amalgamated free product, for example, we will see in §9 that a suitable choice of X leads to the Mayer–Vietoris sequence which we derived earlier from a different point of view (II.7.7 and the exercise of §III.6).

EXERCISES

1. Consider the “left-hand edge” $E'_{0,*}$ of the spectral sequence 7.7. Since d' is zero on this edge for $r \geq 1$, the spectral sequence gives us maps

$$\bigoplus_{r \in \Sigma_0} H_*(G_r, M) = E'_{0,*} \rightarrow E_{0,*}^\infty = \text{Gr}_0 H_*^G(X, M) \hookrightarrow H_*^G(X, M).$$

[We have written $H_*(G_v, M)$ here instead of $H_*(G_v, M_v)$ because a zero-cell has a unique orientation, so that M_v is canonically isomorphic to M as a G_v -module.] The composite

$$\bigoplus_{v \in \Sigma_0} H_*(G_v, M) \rightarrow H_*^G(X, M)$$

is one of the two “edge homomorphisms” associated to the spectral sequence. Show that its restriction to $H_*(G_v, M)$ is the map

$$H_*(G_v, M) = H_*^{G_v}(v, M) \rightarrow H_*^G(X, M)$$

induced by the inclusion $(G_v, v) \hookrightarrow (G, X)$. [Hint: This can be done by straightforward (but tedious) definition-checking. It is easier, however, to use a naturality argument to reduce to the case where X consists of the single vertex v , in which case the definition-checking is trivial.]

2. Let X be a G -complex such that for each cell σ of X , the isotropy group G_σ fixes σ pointwise. In this case it is easy to see that the orbit space X/G inherits a CW -structure. If, in addition, each G_σ is finite, show that

$$H_*^G(X, \mathbb{Q}) \approx H_*(X/G, \mathbb{Q}).$$

Deduce, in particular, that

$$H_*(G, \mathbb{Q}) \approx H_*(X/G, \mathbb{Q})$$

if X is contractible. [Remark: The hypothesis that G_σ fixes σ pointwise is not very restrictive in practice. In the case of a simplicial action, for example, it can always be achieved by passage to the barycentric subdivision.]

3. Show that $H_*^G(X) \approx H_*(X \times^G E)$, where E is a free, contractible G -complex (i.e., the universal cover of a $K(G, 1)$) and $X \times^G E$ denotes $X \times E$ modulo the diagonal G -action. [Hint: There is a G -map $X \times E \rightarrow X$ which is a homotopy equivalence, and G acts freely on $X \times E$.]

Remark. The isomorphism of exercise 3 is often taken as a definition. From this point of view one obtains a spectral sequence of the form 7.2 by noting that there is a fiber bundle $X \times^G E \rightarrow K(G, 1)$ with fiber X .

4. (a) Let X be a G -space (not necessarily a G -complex). Use singular chains to define equivariant singular homology $H_*^G(X)$ and construct a spectral sequence of the form 7.2.

(b) Suppose that G acts freely on X and that the projection $X \rightarrow X/G$ is a covering map. (It is then a regular G -cover in the sense of the appendix to Chapter I.) Prove that $H_*^G(X) \approx H_*(X/G)$. [Hint: 5.5.] Deduce that there is a Cartan–Leray spectral sequence (as in 7.9) in singular homology.

*(c) Suppose $X = S^{2k-1}$ in (b). Prove that G has periodic cohomology of period $2k$. [Method 1: X/G is a finite dimensional manifold and hence has vanishing homology in high dimensions, with arbitrary local coefficients. The differential in the spectral sequence therefore yields $H_i(G, M) \approx H_{i+2k}(G, M)$ for $i \gg 0$. Method 2: We have a sphere bundle $S^{2k-1} \times^G E \rightarrow K(G, 1)$ (with E as in exercise 3) whose total space is homotopy equivalent to S^{2k-1}/G . Up to homotopy, then, the inclusion of the fiber in the total space is a map of degree $|G|$ of orientable $(2k-1)$ -manifolds. The Gysin sequence now shows that the Euler class of the sphere bundle is an element of order $|G|$ in $H^{2k}(G, \mathbb{Z})$, so the result follows from VI.9.1(iv). See Jackowski [1978] for further results obtainable from the study of sphere bundles over $K(G, 1)$.]

5. Let X be a G -complex. If N is a normal subgroup of G which acts freely on X , show that $H_*^G(X) \approx H_*^{G/N}(X/N)$ with any G/N -module of coefficients. [Hint: Apply 7.7 to (G, X) and $(G/N, X/N)$.]
6. Let Y be a connected CW -complex with $\pi_i Y = 0$ for $1 < i < n$ as in II.5.2. Deduce from the Cartan–Leray spectral sequence of the universal cover of Y that there is a 5-term exact sequence

$$H_{n+1} Y \rightarrow H_{n+1} \pi \rightarrow (\pi_n Y)_\pi \rightarrow H_n Y \rightarrow H_n \pi \rightarrow 0,$$

where $\pi = \pi_1 Y$.

8 Computation of d^1

Let X be a G -complex and M a G -module. In this section we will compute the differential d^1 in the spectral sequence 7.7.

If σ is a p -cell of X and τ is a $(p-1)$ -cell, we denote by $\partial_{\sigma\tau}: M_\sigma \rightarrow M_\tau$ the (σ, τ) -component of the boundary operator $\partial: C_p(X, M) \rightarrow C_{p-1}(X, M)$

(cf. 7.4). Let $\mathcal{F}_\sigma = \{\tau: \partial_{\sigma\tau} \neq 0\}$. This is a finite set of $(p-1)$ -cells and is G_σ -invariant. Writing $G_{\sigma\tau} = G_\sigma \cap G_\tau$, we conclude that $(G_\sigma: G_{\sigma\tau}) < \infty$ for $\tau \in \mathcal{F}_\sigma$. There is therefore a transfer map

$$t_{\sigma\tau}: H_*(G_\sigma, M_\sigma) \rightarrow H_*(G_{\sigma\tau}, M_\sigma).$$

Next, note that $\partial_{\sigma\tau}: M_\sigma \rightarrow M_\tau$ is a $G_{\sigma\tau}$ -map, since ∂ is a G -map; so $\partial_{\sigma\tau}$ and the inclusion $G_{\sigma\tau} \hookrightarrow G_\tau$ induce a map

$$u_{\sigma\tau}: H_*(G_{\sigma\tau}, M_\sigma) \rightarrow H_*(G_\tau, M_\tau).$$

Finally, let $\tau_0 \in \Sigma_{p-1}$ be the representative of the G -orbit of τ , and choose $g(\tau) \in G$ such that $g(\tau)\tau = \tau_0$. Then the action of $g(\tau)$ on $C_{p-1}(X, M)$ gives an isomorphism $M_\tau \rightarrow M_{\tau_0}$ compatible with the conjugation isomorphism $G_\tau \rightarrow G_{\tau_0}$ given by $g \mapsto g(\tau)g g(\tau)^{-1}$; thus there is an induced isomorphism

$$v_\tau: H_*(G_\tau, M_\tau) \rightarrow H_*(G_{\tau_0}, M_{\tau_0}).$$

[As the notation suggests, one can show that v_τ depends only on τ , and not on the choice of $g(\tau)$; but we will not need to know this].

We can now define a map

$$\varphi: \bigoplus_{\sigma \in \Sigma_p} H_*(G_\sigma, M_\sigma) \rightarrow \bigoplus_{\tau \in \Sigma_{p-1}} H_*(G_\tau, M_\tau)$$

by

$$\varphi|H_*(G_\sigma, M_\sigma) = \sum_{\tau \in \mathcal{F}_\sigma} v_\tau u_{\sigma\tau} t_{\sigma\tau},$$

where $\mathcal{F}_\sigma$ is a set of representatives for $\mathcal{F}_\sigma/G_\sigma$.

(8.1) Proposition. *Up to sign, the map φ is the differential $d^1: E_{p,*}^1 \rightarrow E_{p-1,*}^1$ in 7.7.*

PROOF. What we must prove is that the diagram

$$\begin{array}{ccc} \bigoplus_{\sigma \in \Sigma_p} H_*(G_\sigma, M_\sigma) & \xrightarrow{\varphi} & \bigoplus_{\tau \in \Sigma_{p-1}} H_*(G_\tau, M_\tau) \\ \downarrow \approx & & \downarrow \approx \\ H_*(G, C_p(X, M)) & \xrightarrow{H_*(G, \partial)} & H_*(G, C_{p-1}(X, M)) \end{array}$$

commutes, where the vertical isomorphisms come from 7.6. We will in fact prove commutativity of the corresponding diagram of chain complexes. More precisely, if F is a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}G$, we will compute φ on the chain level as a map (still denoted φ)

$$\bigoplus_{\sigma \in \Sigma_p} F \otimes_{G_\sigma} M_\sigma \rightarrow \bigoplus_{\tau \in \Sigma_{p-1}} F \otimes_{G_\tau} M_\tau,$$

and we will prove that the diagram

$$\begin{array}{ccc}
 \bigoplus_{\sigma \in \Sigma_p} F \otimes_{G_\sigma} M_\sigma & \xrightarrow{\varphi} & \bigoplus_{\tau \in \Sigma_{p-1}} F \otimes_{G_\tau} M_\tau \\
 \theta \downarrow & & \downarrow \psi \\
 F \otimes_G C_p(X, M) & \xrightarrow{F \otimes \partial} & F \otimes_G C_{p-1}(X, M)
 \end{array}$$

commutes. Here θ and ψ are the chain isomorphisms underlying 7.6; explicitly, a glance at the proof of Shapiro's lemma (III.6.2) shows that θ and ψ are simply the maps induced by the inclusions $M_\sigma \hookrightarrow C_p(X) \otimes M$ and $M_\tau \hookrightarrow C_{p-1}(X) \otimes M$.

We begin by computing the transfer map $t_{\sigma\tau}$ on the chain level by means of Definition (C) of §III.9. For $x \in F$ and $m \in M_\sigma$, we find

$$t_{\sigma\tau}(x \otimes m) = \sum_{g \in G_\sigma/G_{\sigma\tau}} g^{-1}x \otimes g^{-1}m \in F \otimes_{G_{\sigma\tau}} M_\sigma.$$

Next, the map $u_{\sigma\tau}$ is given on the chain level by $F \otimes \partial_{\sigma\tau}: F \otimes_{G_{\sigma\tau}} M_\sigma \rightarrow F \otimes_{G_\tau} M_\tau$. Thus

$$\begin{aligned}
 u_{\sigma\tau} t_{\sigma\tau}(x \otimes m) &= \sum_{g \in G_\sigma/G_{\sigma\tau}} g^{-1}x \otimes \partial_{\sigma\tau}(g^{-1}m) \\
 &= \sum_{g \in G_\sigma/G_{\sigma\tau}} g^{-1}x \otimes g^{-1}\partial_{\sigma, g\tau}(m),
 \end{aligned}$$

where the second equality comes from the fact that ∂ is a G -map. We apply now the map v_τ , which can be computed via the "diagonal action" of $g(\tau)$ (cf. §III.8, paragraph preceding III.8.1), and we sum over $\tau \in \mathcal{F}'_\sigma$; this gives

$$\varphi(x \otimes m) = \sum_{\tau \in \mathcal{F}'_\sigma} \sum_{g \in G_\sigma/G_{\sigma\tau}} g(\tau)g^{-1}x \otimes g(\tau)g^{-1}\partial_{\sigma, g\tau}(m).$$

We can now establish the desired commutative diagram of chain complexes. Using the tensor product relations $gu \otimes gv = u \otimes v$ in $- \otimes_G -$, we deduce from the formula for φ above that

$$\psi \varphi(x \otimes m) = \sum_{\tau \in \mathcal{F}'_\sigma} \sum_{g \in G_\sigma/G_{\sigma\tau}} x \otimes \partial_{\sigma, g\tau}(m).$$

But it is clear from the definition of $\mathcal{F}'_\sigma$ that the right-hand side of this equation is simply

$$\sum_{\tau \in \mathcal{F}'_\sigma} x \otimes \partial_{\sigma\tau}(m) = x \otimes \partial m.$$

Thus $\psi \varphi = (F \otimes \partial)\theta$, as required. $\square$

The description of E^1 and d^1 simplifies greatly if X has the property that G_σ fixes σ pointwise for every cell σ , as in exercise 2 of §7. Indeed, one can then orient each cell of X in such a way that the G -action preserves orientations, and such a choice of orientations determines for each σ an isomorphism

$M_\sigma \approx M$ of G_σ -modules. We can therefore write $H_q(G_\sigma, M)$ instead of $H_q(G_\sigma, M_\sigma)$. More significantly, we have $G_{\sigma\tau} = G_\sigma$ for $\tau \in \mathcal{F}_\sigma$, since such a τ is necessarily in the closure of σ and hence is fixed by G_σ . Thus the transfer map $t_{\sigma\tau}$ is the identity map, and also $\mathcal{F}'_\sigma = \mathcal{F}_\sigma$.

Roughly speaking, then, we have a “coefficient system”

$$\mathcal{H}_q = \{H_q(G_\sigma, M)\}$$

on the orbit complex X/G , and (E^1, d^1) is the chain complex of X/G with coefficients in this system. The E^2 -term of the spectral sequence can therefore be thought of as $H_p(X/G, \mathcal{H}_q)$. We will not attempt to make these remarks precise.

9 Example: Amalgamations

Consider an amalgamation $G = H *_A K$, where $A \hookrightarrow H$ and $A \hookrightarrow K$. Let X be the tree associated to G (cf. appendix to Chapter II), and recall that G acts on X with no “inversions,” i.e., no element of G interchanges the vertices of a 1-simplex of X . [Thus we are in the situation described at the end of §8.] There is a single 1-simplex e which maps isomorphically onto the quotient graph X/G , and the isotropy groups of e and its vertices v and w are given by

$$G_v = H, \quad G_w = K, \quad \text{and} \quad G_e = A.$$

We therefore have, for any G -module M , a spectral sequence converging to $H_*(G, M)$ (cf. 7.10), with

$$E_{p,*}^1 = \begin{cases} H_*(H, M) \oplus H_*(K, M) & \text{if } p = 0 \\ H_*(A, M) & \text{if } p = 1 \\ 0 & \text{if } p > 1. \end{cases}$$

(We have taken, here, $\Sigma_0 = \{v, w\}$ and $\Sigma_1 = \{e\}$.) The spectral sequence therefore collapses at E^2 to yield a Mayer–Vietoris sequence

$$(9.1) \quad \cdots \rightarrow H_n(A, M) \xrightarrow{\alpha} H_n(H, M) \oplus H_n(K, M) \xrightarrow{\beta} H_n(G, M) \\ \rightarrow H_{n-1}(A, M) \rightarrow \cdots$$

The map β here is the edge homomorphism discussed in exercise 1 of §7, and hence it is the map induced by the inclusions $H \hookrightarrow G$ and $K \hookrightarrow G$. The map α is the differential d^1 given by 8.1. The latter is particularly easy to apply here because the maps $t_{\sigma\tau}$ and v_τ of §8 are identity maps. Since $\partial_{e,w}: M \rightarrow M$ is id_M and $\partial_{e,v}: M \rightarrow M$ is $-\text{id}_M$, we conclude that α is simply the difference of the maps $H_*(A) \rightarrow H_*(H)$ and $H_*(A) \rightarrow H_*(K)$ induced by the inclusions $A \hookrightarrow H$ and $A \hookrightarrow K$.

Next we wish to briefly describe a generalization of the notion of amalgamation, for which one can also construct a Mayer–Vietoris sequence. By a *graph of groups* we mean a connected graph Y together with (a) groups G_v and G_e , where v (resp. e) ranges over the vertices (resp. edges) of Y , and (b) monomorphisms $\theta_0: G_e \hookrightarrow G_v$ and $\theta_1: G_e \hookrightarrow G_w$ for every edge e , where v and w are the vertices of e . [Note: We allow the possibility that $v = w$, but θ_0 and θ_1 may still differ.] One can then define the *fundamental group* G of the graph of groups by using a suitable notion of “path”: one regards an element of G_v as a path from v to v , and one imposes the relations

$$\theta_0(a)e = e\theta_1(a)$$

for every $a \in G_e$, i.e., one identifies the composite paths

$$\theta_0(a) \begin{array}{c} \circlearrowleft \\ \text{---} t \end{array} \xrightarrow{e} \text{---} w \quad \text{and} \quad \text{---} v \xrightarrow{e} \text{---} w \begin{array}{c} \circlearrowright \\ \text{---} w \end{array} \theta_1(a)$$

from v to w . (See Serre [1977a] for more details.)

For example, the amalgam $H *_A K$ is the fundamental group of the graph

$$\begin{array}{ccc} & A & \\ \circ & \text{---} & \circ \\ H & & K \end{array}$$

where we have labelled the vertices and edge with the associated groups. (In this case θ_0 and θ_1 are the inclusions $A \hookrightarrow H$, $A \hookrightarrow K$.)

Another important group theoretic construction which fits into this framework is the *HNN extension* $H *_A$. Here we are given a group H , a subgroup A , and a monomorphism $\theta: A \rightarrow H$, and $H *_A$ is obtained by adjoining an element t to H subject to the relations $t^{-1}at = \theta(a)$ ($a \in A$). This group can be realized as the fundamental group of the graph

$$\begin{array}{c} A \\ \circlearrowleft \\ \circ \end{array} H$$

with one vertex and one edge, where θ_0 and θ_1 are the inclusion $i: A \hookrightarrow H$ and the given map $\theta: A \rightarrow H$.

Just as in the case of amalgamation, there is a tree X associated to a graph of groups, and the fundamental group G acts on X without inversion. The original graph Y is the orbit graph X/G , and the groups G_v and G_e are the isotropy subgroups of G at suitable liftings to X of the vertices and edges of Y . Consequently, one obtains from 7.10 a Mayer–Vietoris sequence

$$\begin{aligned} (9.2) \quad \cdots \rightarrow \bigoplus_{e \in Y_1} H_n(G_e, M) &\rightarrow \bigoplus_{v \in Y_0} H_n(G_v, M) \rightarrow H_n(G, M) \\ &\rightarrow \bigoplus_{e \in Y_1} H_{n-1}(G_e, M) \rightarrow \cdots, \end{aligned}$$

generalizing 9.1 (Here Y_i is the set of i -cells of Y .) This exact sequence was first written down explicitly by Chiswell [1976a].

[This exact sequence can, of course, be derived without the use of spectral sequences. Namely, one argues as in the exercise of §III.6, using the exact sequence of G -modules

$$(9.3) \quad 0 \rightarrow \bigoplus_{e \in Y_1} \mathbb{Z}[G/G_e] \rightarrow \bigoplus_{v \in Y_0} \mathbb{Z}[G/G_v] \rightarrow \mathbb{Z} \rightarrow 0$$

given by the augmented chain complex of the tree X .]

Consider, for example, the HNN extension $G = H *_A$ described above. The sequence 9.2 then has the form

$$(9.4) \quad \cdots \rightarrow H_n(A, M) \xrightarrow{\alpha} H_n(H, M) \xrightarrow{\beta} H_n(G, M) \rightarrow H_{n-1}(A, M) \rightarrow \cdots$$

As in 9.1, β is induced by the inclusion $H \rightarrow G$. To evaluate $\alpha = d^1$, we need to know the following fact about the tree X : the unique edge of Y lifts to an edge e of X whose vertices have the form v, tv . Taking $\Sigma_0 = \{v\}$ and $\Sigma_1 = \{e\}$ in the notation of §8, we find that v_τ ($\tau = tv$) is the conjugation isomorphism $c(t^{-1})_*: H_*(tHt^{-1}, M) \rightarrow H_*(H, M)$. Since the composite

$$(A, M) \rightarrow (tHt^{-1}, M) \xrightarrow{\alpha(t^{-1})} (H, M)$$

is equal to (θ, t^{-1}) , it follows that $\alpha = (\theta, t^{-1})_* - i_*$.

10 Equivariant Tate Cohomology

Recall from §7 that $\hat{H}_G^*(X, M)$ is defined if G is finite and the G -complex X is finite dimensional. This theory was introduced by Swan [1960a], who in fact constructed a *topological* version of the theory (i.e., X is not required to be a CW -complex).

The usefulness of equivariant Tate cohomology comes from the fact that it provides a machine for systematically ignoring free actions:

(10.1) Proposition. *Let Y be a G -invariant subcomplex of X such that the isotropy group G_σ is trivial for every cell σ of X that is not in Y . Then the inclusion $Y \hookrightarrow X$ induces an isomorphism $\hat{H}_G^*(X, M) \xrightarrow{\cong} \hat{H}_G^*(Y, M)$.*

PROOF. Consider the Tate cohomology version of the spectral sequence 7.7 for (G, X) and (G, Y) . By hypothesis, $\hat{H}^*(G_\sigma, M) = 0$ if σ is cell of $X - Y$. Thus the inclusion $Y \hookrightarrow X$ induces an isomorphism of spectral sequences and hence an isomorphism of the abutments $\hat{H}_G^*(X, M)$ and $\hat{H}_G^*(Y, M)$. $\square$

As a simple illustration of 10.1, we prove the following result (cf. §VI.8, exercise 4):

(10.2) Proposition. *Let G be a finite group which admits a finite dimensional free G -complex X such that $H_*(X) \approx H_*(S^{2k-1})$. Then G has periodic cohomology of period $2k$.*

PROOF. By 10.1 (with $Y = \emptyset$), $\hat{H}_G^*(X, M) = 0$. On the other hand, we have a spectral sequence (cf. 7.2)

$$(10.3) \quad E_2^{p,q} = \hat{H}^p(G, H^q(X, M)) \Rightarrow \hat{H}_G^{p+q}(X, M).$$

Since the spectral sequence is concentrated on the horizontal lines $q = 0$ and $q = 2k - 1$, it follows that the differential d_{2k} is an isomorphism

$$(10.4) \quad \hat{H}^p(G, H^{2k-1}(X, M)) \xrightarrow{\sim} \hat{H}^{p+2k}(G, M).$$

In view of the universal coefficient isomorphism

$$H^{2k-1}(X, M) \approx \text{Hom}(H_{2k-1}X, M),$$

which is natural and hence an isomorphism of G -modules, the proof will be complete if we show that G acts trivially on the infinite cyclic group $H_{2k-1}X$. This can be deduced from a general Lefschetz fixed-point theorem (cf. §IX.5, exercise 1), but in the present situation there is a much easier proof. It suffices to prove that every cyclic subgroup of G acts trivially on $H_{2k-1}X$, so we are immediately reduced to the case where G is cyclic (and non-trivial). In this case we apply 10.4 with $M = \mathbb{Z}$ and $p = 0$ to conclude that

$$\hat{H}^0(G, H^{2k-1}(X, \mathbb{Z})) \approx \hat{H}^{2k}(G, \mathbb{Z}) \approx \mathbb{Z}/|G| \cdot \mathbb{Z}.$$

But this implies that $H^{2k-1}(X, \mathbb{Z})^G \neq 0$, which can only happen if G acts trivially on $H_{2k-1}X$. $\square$

Remark. The advantage of this spectral sequence proof of 10.2 is that it applies verbatim to the topological situation, where X is not assumed to be a CW-complex, cf. Swan [1960a].

As another application of equivariant Tate cohomology theory (also due to Swan), we will prove some classical results of fixed-point theory. (See Bredon [1972] or Borel et al [1960] for the standard proofs of these results, based on "Smith theory.")

(10.5) Theorem. *Let X be a finite-dimensional G -complex, where G has prime order p . Assume that the fixed-point set X^G is a subcomplex.*

- (a) *If $H^*(X, \mathbb{Z}_p)$ is finitely generated, then so is $H^*(X^G, \mathbb{Z}_p)$.*
- (b) *If X is acyclic mod p (i.e., $H^*(X, \mathbb{Z}_p) \approx H^*(\text{pt.}, \mathbb{Z}_p)$), then so is X^G .*
- (c) *If X is a cohomology sphere mod p (i.e., $H^*(X, \mathbb{Z}_p) \approx H^*(S^n, \mathbb{Z}_p)$ for some $n \geq 0$), then so is X^G , provided $X^G \neq \emptyset$.*

(The hypothesis that X^G is a subcomplex holds, for example, if X satisfies the hypothesis of exercise 2 of §7.)

PROOF. We have

$$\hat{H}_G^*(X, \mathbb{Z}_p) \approx \hat{H}_G^*(X^G, \mathbb{Z}_p)$$

by 10.1. On the other hand, since G acts trivially on X^G , we have a Künneth isomorphism

$$\hat{H}_G^*(X^G, \mathbb{Z}_p) \approx \hat{H}^*(G, \mathbb{Z}_p) \otimes_{\mathbb{Z}_p} H^*(X^G, \mathbb{Z}_p),$$

cf. 5.4. Since $\dim_{\mathbb{Z}_p} \hat{H}^n(G, \mathbb{Z}_p) = 1$ for all $n \in \mathbb{Z}$, it follows that

$$(10.6) \quad \dim_{\mathbb{Z}_p} \hat{H}_G^n(X, \mathbb{Z}_p) = \sum_{i \geq 0} \dim_{\mathbb{Z}_p} H^i(X^G, \mathbb{Z}_p)$$

for all n . Assuming now that $H^*(X, \mathbb{Z}_p)$ is finitely generated, the spectral sequence 10.3 (with $M = \mathbb{Z}_p$) implies that $\hat{H}_G^n(X, \mathbb{Z}_p)$ is finitely generated, so (a) follows from 10.6. Similarly, if X is acyclic mod p then the left side of 10.6 is equal to 1, hence so is the right side; this proves (b). The proof of (c) is identical to that of (b), except that one uses relative cohomology groups of the pair (X, v) , where v is some vertex of X^G . $\square$

EXERCISES

1. Where did the proof of 10.5 use the assumption that $|G| = p$? [There are two places.]
2. Extend 10.5 to the case where G is an arbitrary p -group and X^H is a subcomplex of X for every $H \subseteq G$. [Hint: Use induction on $|G|$, noting that $X^G = (X^N)^{G/N}$ if $N \triangleleft G$.]
3. Using the method of proof of 10.2, show that if X is a finite-dimensional free G -complex (G finite) with $H_*(X) \approx H_*(S^{2k})$, then every non-trivial element of G acts non-trivially on $H_{2k}(X)$, and hence $|G| \leq 2$.

Finiteness Conditions

1 Introduction

Recall that the definition of $H_*(G, M)$ and $H^*(G, M)$ allows us to choose an arbitrary projective resolution $P = (P_i)_{i \geq 0}$ of $\mathbb{Z}$ over $\mathbb{Z}G$. Similarly, if we wish to take the topological point of view, then we can compute $H_*(G, M)$ and $H^*(G, M)$ in terms of an arbitrary $K(G, 1)$ -complex Y . Since we have this freedom of choice, it is reasonable to try to choose P (or Y) to be as “small” as possible, and this leads to various finiteness conditions on G .

For example, if we interpret “small” in terms of the length of P (or the dimension of Y), then we are led to the notion of *cohomological dimension*. Or if we interpret small to mean that each P_i should be finitely generated (or that Y should have only finitely many cells), then we are led to the so-called “*FP*” and “*FL*” conditions.

Our goal in this chapter is to introduce these and related finiteness conditions and to give some examples. Our treatment will by no means be complete; for the most part we will give complete proofs only for those results which will be needed in Chapters IX and X. See Bieri [1976] and Serre [1971, 1979] for a much more thorough treatment of the subject and a guide to the literature. See also Wall [1979] for a list of open questions concerning finiteness conditions.

Finally, a word about notation: In the theory of discrete subgroups of Lie groups, which is the source of many of our examples, it is customary to denote the Lie group by G and the discrete subgroup by Γ . In order to be consistent with this, we will use the letter “ Γ ” from now on (instead of “ G ”) to denote a typical abstract group.

2 Cohomological Dimension

We begin with a basic lemma which characterizes projective dimension in terms of the vanishing of cohomology functors. Recall first (cf. §VI.8) that if R is a ring, M is an R -module, and n is a non-negative integer, then $\text{proj dim}_R M \leq n$ if and only if M admits a projective resolution

$$0 \rightarrow P_n \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$$

of length n . Recall also (cf. §III.2) that the Ext functors are defined by

$$\text{Ext}_R^i(M, -) = H^i(\mathcal{H}om_R(P, -)),$$

where P is a projective resolution of M . In particular,

$$\text{Ext}_{\mathbb{Z}\Gamma}^i(\mathbb{Z}, -) = H^i(\Gamma, -).$$

(2.1) Lemma. *The following conditions are equivalent:*

- (i) $\text{proj dim}_R M \leq n$.
- (ii) $\text{Ext}_R^i(M, -) = 0$ for $i > n$.
- (iii) $\text{Ext}_R^{n+1}(M, -) = 0$.
- (iv) *If $0 \rightarrow K \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0$ is any exact sequence of R -modules with each P_i projective, then K is projective.*

PROOF. It is obvious that (iv) $\Rightarrow$ (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii), so we need only prove (iii) $\Rightarrow$ (iv). Given a partial resolution as in (iv), complete it arbitrarily to a projective resolution

$$\begin{array}{ccccccc} \cdots & \rightarrow & P_{n+1} & \rightarrow & P_n & \rightarrow & P_{n-1} \rightarrow \cdots \rightarrow P_0 \rightarrow M \rightarrow 0. \\ & & \searrow & & \nearrow & & \searrow & & \nearrow \\ & & & L & & & & K & \end{array}$$

For any R -module N , an $(n+1)$ -cocycle in $\mathcal{H}om_R(P, N)$ is a map $P_{n+1} \rightarrow N$ whose composition with $P_{n+2} \rightarrow P_{n+1}$ is zero. Such a cocycle, therefore, can be regarded as a map $\varphi: L \rightarrow N$. The cocycle is a coboundary if and only if φ extends to a map $P_n \rightarrow N$. Thus (iii) implies that every map on L extends to P_n . In particular, the identity map on L extends to P_n , so $P_n \approx L \oplus K$ and hence K is projective. $\square$

The implication (i) $\Rightarrow$ (iv) of 2.1 is very useful. It shows that if there exist projective resolutions of length n , then we don't have to be clever to find one—any partial resolution of length $n-1$ can be completed to a projective resolution of length n . We will give another proof of this important fact later (remark 3 following Lemma 4.4).

We now specialize to the case $R = \mathbb{Z}\Gamma$, $M = \mathbb{Z}$. The *cohomological dimension* of Γ , denoted $\text{cd } \Gamma$, is defined to be the smallest integer n such that

the conditions of the lemma hold, provided there exist such integers n ; otherwise we set $\text{cd } \Gamma = \infty$. Thus

$$\begin{aligned}\text{cd } \Gamma &= \text{proj dim}_{\mathbb{Z}\Gamma} \mathbb{Z} \\ &= \inf\{n: \mathbb{Z} \text{ admits a projective resolution of length } n\} \\ &= \inf\{n: H^i(\Gamma, -) = 0 \text{ for } i > n\} \\ &= \sup\{n: H^n(\Gamma, M) \neq 0 \text{ for some } \Gamma\text{-module } M\}.\end{aligned}$$

There is an obvious topological analogue of $\text{cd } \Gamma$: The *geometric dimension* of Γ , denoted $\text{geom dim } \Gamma$, is defined to be the minimal dimension of a $K(\Gamma, 1)$ -complex. Since the cellular chain complex of the universal cover of a $K(\Gamma, 1)$ complex Y yields a free resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ (of length equal to the dimension of Y), we clearly have:

(2.2) Proposition. $\text{cd } \Gamma \leq \text{geom dim } \Gamma$.

We will return to this in §7, where we will prove that equality usually holds.

EXAMPLES

1. $\text{cd } \Gamma = 0$ if and only if Γ is the trivial group (cf. exercise 1 below).
2. If Γ is free and non-trivial then $\text{cd } \Gamma = 1$ (cf. I.4.3). Conversely, a deep theorem of Stallings [1968] and Swan [1969] says that every group of cohomological dimension 1 is free. In view of Chapter IV, this result can be restated as follows: If Γ is a group which admits no non-split extension with abelian kernel, then Γ admits no non-split extension at all.
3. Let Γ be the fundamental group of a connected closed surface Y other than S^2 or P^2 . Then Y is a 2-dimensional $K(\Gamma, 1)$ (cf. §II.4, example 2), so $\text{cd } \Gamma \leq 2$. And $H^2(\Gamma, \mathbb{Z}_2) \approx H^2(Y, \mathbb{Z}_2) \neq 0$, so $\text{cd } \Gamma = 2$.
4. More generally, suppose Γ is a one-relator group whose relator is not a proper power. Then $\text{cd } \Gamma \leq 2$ by Lyndon's theorem which we quoted in §II.4, example 3.
5. If $\Gamma = \mathbb{Z}^n$ then the n -dimensional torus $Y = S^1 \times \cdots \times S^1$ is a $K(\Gamma, 1)$ with $H^n(Y, \mathbb{Z}) \approx \mathbb{Z} \neq 0$, hence $\text{cd } \Gamma = n$.
6. Let Γ be the group of 3×3 strictly upper triangular matrices with integral entries:

$$\Gamma = \begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix}$$

We will show that $\text{cd } \Gamma = 3$. Let Γ' be the central subgroup

$$\begin{pmatrix} 1 & 0 & * \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

of Γ , and note that the quotient $\Gamma'' = \Gamma/\Gamma'$ is free abelian of rank 2. Consider the Hochschild–Serre spectral sequence

$$E_2^{p,q} = H^p(\Gamma'', H^q(\Gamma', M)) \Rightarrow H^{p+q}(\Gamma, M),$$

where M is an arbitrary Γ -module. Example 5 shows that this spectral sequence is concentrated in the rectangle $0 \leq p \leq 2, 0 \leq q \leq 1$. Consequently, $H^i(\Gamma, M) = 0$ for $i > 3$, so $\text{cd } \Gamma \leq 3$. Moreover, since the differential d_r maps $E_r^{p,q}$ to $E_r^{p+r, q-r+1}$, there can be no non-zero differentials involving the upper right-hand corner $E_r^{2,1}$. This being the only non-trivial term of total degree 3, it follows that

$$H^3(\Gamma, M) = E_\infty^{2,1} = E_2^{2,1} = H^2(\Gamma'', H^1(\Gamma', M)).$$

In particular, taking $M = \mathbb{Z}$ and recalling that Γ' is central in Γ , we see that $H^3(\Gamma, \mathbb{Z}) \approx \mathbb{Z}$ (cf. exercise 1 of §III.8); hence $\text{cd } \Gamma = 3$.

7. More generally, let Γ be an arbitrary finitely generated, torsion-free, nilpotent group. One can show that Γ admits a central series

$$\Gamma = \Gamma_0 \supset \Gamma_1 \supset \cdots \supset \Gamma_n = \{1\}$$

with free abelian quotients Γ_i/Γ_{i+1} . The sum of the ranks of these quotients is independent of the choice of central series and is called the *rank* (or *Hirsch number*) of Γ . Arguing as in example 6, one finds

$$\text{cd } \Gamma = \text{rank } \Gamma.$$

Details are omitted. See Bieri [1976], §7.3, and Gruenberg [1970], §8.8, for more details and for further results of this type. See also example 2 of §9 below for an indication of a different proof of this result.

The rest of this section will be devoted to some elementary properties of cohomological dimension.

(2.3) Proposition. *If $\text{cd } \Gamma < \infty$ then*

$$\text{cd } \Gamma = \sup\{n: H^n(\Gamma, F) \neq 0 \text{ for some free } \mathbb{Z}\Gamma\text{-module } F\}.$$

PROOF. Let $n = \text{cd } \Gamma$. In view of the long exact cohomology sequence (III.6.1(ii')), the functor $H^n(\Gamma, -)$ is right exact. Since $H^n(\Gamma, M) \neq 0$ for some M , it follows that $H^n(\Gamma, F) \neq 0$ for any free module F which maps onto M . $\square$

(2.4) Proposition.

(a) If $\Gamma' \subset \Gamma$ then

$$\text{cd } \Gamma' \leq \text{cd } \Gamma;$$

equality holds if $\text{cd } \Gamma < \infty$ and $(\Gamma: \Gamma') < \infty$.

(b) If $1 \rightarrow \Gamma' \rightarrow \Gamma \rightarrow \Gamma'' \rightarrow 1$ is a short exact sequence of groups, then

$$\text{cd } \Gamma \leq \text{cd } \Gamma' + \text{cd } \Gamma''.$$

(c) If $\Gamma = \Gamma_1 *_A \Gamma_2$ (where $A \hookrightarrow \Gamma_i$), then

$$\text{cd } \Gamma \leq \max\{\text{cd } \Gamma_1, \text{cd } \Gamma_2, 1 + \text{cd } A\}.$$

PROOF. The inequality in (a) follows immediately from Shapiro's lemma or, alternatively, from the fact that a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ can also be regarded as a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma'$. To prove the second part of (a), suppose $\text{cd } \Gamma = n < \infty$. By 2.3 there is a free $\mathbb{Z}\Gamma$ -module F with $H^n(\Gamma, F) \neq 0$. If F' is a free $\mathbb{Z}\Gamma'$ -module of the same rank, then $F \approx \text{Ind}_{\Gamma'}^{\Gamma} F'$, so Shapiro's lemma yields $H^n(\Gamma', F') \approx H^n(\Gamma, F) \neq 0$. Thus $\text{cd } \Gamma' \geq n$, whence (a). [Exercise: Where did we use the hypothesis that $(\Gamma: \Gamma') < \infty$?] (b) is an immediate consequence of the Hochschild–Serre spectral sequence, as in Example 6 above. Finally, (c) follows from the cohomology version of the Mayer–Vietoris sequence (VII.9.1 or exercise of §III.6). $\square$

(2.5) Corollary. If $\text{cd } \Gamma < \infty$ then Γ is torsion-free.

PROOF. If Γ is not torsion-free then Γ contains a nontrivial finite cyclic subgroup Γ' . Such a Γ' has $\text{cd } \Gamma' = \infty$, since $H^{2k}(\Gamma', \mathbb{Z}) \neq 0$ for all k (cf. §III.1, Example 2) so 2.4a implies that $\text{cd } \Gamma = \infty$. $\square$

If $\Gamma' \subset \Gamma$ and $(\Gamma: \Gamma') < \infty$, then 2.4a shows that $\text{cd } \Gamma' = \text{cd } \Gamma$ unless the following occurs:

$$(*) \quad \text{cd } \Gamma = \infty \quad \text{and} \quad \text{cd } \Gamma' < \infty.$$

Easy examples show that $(*)$ can in fact occur; e.g., take Γ of order 2 and $\Gamma' = \{1\}$. More generally, if Γ is any group with torsion, then $\text{cd } \Gamma = \infty$ by 2.5, but Γ may very well have torsion-free subgroups Γ' of finite index with $\text{cd } \Gamma' < \infty$. We will prove in the next section a theorem of Serre which says that *all* examples of $(*)$ are of this type, i.e., that $(*)$ cannot occur if Γ is torsion-free.

Our last result shows that, as far as cohomological dimension is concerned, we never need to use projective resolutions which are not free:

(2.6) Proposition. For any group Γ there is a free resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of length equal to $\text{cd } \Gamma$.

The proof requires the following “Eilenberg trick”:

(2.7) Lemma. *If P is a projective module over an arbitrary ring R , then there is a free module F such that $P \oplus F \approx F$.*

(Warning: F will be of infinite rank, in general, even if P is finitely generated.)

PROOF. Since P is projective there is a module Q such that $P \oplus Q$ is free. Let F be the countable direct sum

$$(P \oplus Q) \oplus (P \oplus Q) \oplus \cdots.$$

Then F is free, being a direct sum of free modules. But F can also be described as the sum of a countable number of copies of P and a countable number of copies of Q . Adding one more copy of P does not change this, so $P \oplus F \approx F$. $\square$

PROOF OF 2.6. Let $n = \text{cd } \Gamma$. We may assume $0 < n < \infty$. Choose a partial free resolution

$$F_{n-1} \xrightarrow{\partial} \cdots \rightarrow F_0 \rightarrow \mathbb{Z} \rightarrow 0$$

of length $n - 1$ and let $P = \ker\{\partial: F_{n-1} \rightarrow F_{n-2}\}$. (Here we set $F_{-1} = \mathbb{Z}$ if $n = 1$.) Then P is projective by 2.1, so 2.7 gives us a free module F such that $P \oplus F$ is free. Thus if we replace F_{n-1} by $F_{n-1} \oplus F$ and set $\partial|_F = 0$, we obtain a partial free resolution of length $n - 1$ with $\ker \partial$ free, whence the proposition. $\square$

EXERCISES

1. Prove that the trivial group is the only group of cohomological dimension zero. [Hint: $\text{cd } \Gamma = 0 \Leftrightarrow \mathbb{Z}$ is a projective $\mathbb{Z}\Gamma$ -module $\Leftrightarrow$ the canonical surjection $\varepsilon: \mathbb{Z}\Gamma \rightarrow \mathbb{Z}$ splits.]
2. Give an example to show that the hypothesis $\text{cd } \Gamma < \infty$ is necessary in 2.3.
3. (a) Show that $\text{geom dim } \Gamma_1 *_A \Gamma_2 \leq \max\{\text{geom dim } \Gamma_1, \text{geom dim } \Gamma_2, 1 + \text{geom dim } A\}$.
 * (b) If $1 \rightarrow \Gamma' \rightarrow \Gamma \rightarrow \Gamma'' \rightarrow 1$ is exact, show that $\text{geom dim } \Gamma \leq \text{geom dim } \Gamma' + \text{geom dim } \Gamma''$.
4. Prove the following generalization of 2.4c: If Γ is an arbitrary group and X is an acyclic Γ -complex, then

$$\text{cd } \Gamma \leq \sup_{\sigma} \{\text{cd } \Gamma_{\sigma} + \dim \sigma\},$$

where σ ranges over a set of representatives for the cells of $X \bmod \Gamma$. (Taking X to be the tree associated to an amalgam, we recover 2.4c.) [Hint: Use the cohomology version of the spectral sequence VII.7.10. Or see Serre [1971] for an alternative proof.]

5. The purpose of this exercise is to give a cohomological criterion for finite-dimensionality (up to homotopy) of an arbitrary chain complex. In case the chain complex is a projective resolution, this result ((b) below) reduces to the equivalence (i) $\Leftrightarrow$ (ii) $\Leftrightarrow$ (iv) of Lemma 2.1. Let C be a chain complex over an arbitrary ring R .

(a) For any integer n , prove that the following two conditions are equivalent:

- (i) $H^{n+1}(\mathcal{H}om_R(C, M)) = 0$ for all R -modules M .
- (ii) $H_{n+1}C = 0$ and the module B_n of n -boundaries is a direct summand of C_n .

[Hint: Examine cocycles and coboundaries in $\mathcal{H}om_R(C, M)$ as in the proof of 2.1.]

(b) For any integer n , prove that the following three conditions are equivalent:

- (i) C is homotopy equivalent to a complex C' such that $C'_i = 0$ for $i > n$.
- (ii) $H^i(\mathcal{H}om_R(C, M)) = 0$ for all $i > n$ and all R -modules M .
- (iii) If C' is the quotient of C defined by

$$C'_i = \begin{cases} C_i & \text{if } i < n \\ C_n/B_n & \text{if } i = n \\ 0 & \text{if } i > n, \end{cases}$$

then the quotient map $C \rightarrow C'$ is a homotopy equivalence.

If these conditions hold and C is a complex of *projective* R -modules, prove that the complex C' in (iii) is also a complex of projectives.

[Hint for (ii) $\Rightarrow$ (iii): If (ii) holds, deduce from (a) that $C \approx C' \oplus C''$, where C'' is contractible.]

- *6. Suppose $\text{cd } \Gamma = n < \infty$, and let $\Gamma' \subseteq \Gamma$ be a subgroup of finite index.

(a) For any Γ -module M , show that the transfer map $\text{tr}: H^n(\Gamma', M) \rightarrow H^n(\Gamma, M)$ is surjective. [Hint: Compute tr on the chain level, using a projective resolution of $\mathbb{Z}$ over $\mathbb{Z}\Gamma$ of length n ; note that the chain map is surjective. Alternatively, see Serre [1971], 1.3, Lemme 2.]

(b) Suppose that Γ' is normal in Γ . If M is a Γ -module and d is an integer such that $H^i(\Gamma', M) = 0$ for $i > d$, show that $H^i(\Gamma, M) = 0$ for $i > d$ and that tr induces an isomorphism

$$H^d(\Gamma', M)_{\Gamma/\Gamma'} \xrightarrow{\sim} H^d(\Gamma, M).$$

In particular, this holds with $d = n$ for any M . [See Brown–Kahn [1977], Prop. 1.2.]

7. (a) Show that induced Γ -modules $\mathbb{Z}\Gamma \otimes A$ have projective dimension ≤ 1 .

(b) If $\text{proj dim}_R M \leq n$, show that $\text{proj dim}_R M' \leq n$ for any direct summand M' of M . [Hint: 2.1.]

(c) Deduce from (a) and (b) the following result (which is also an immediate consequence of VI.8.12): If Γ is finite and M is a Γ -module in which $|\Gamma|$ is invertible, then $\text{proj dim}_{\mathbb{Z}\Gamma} M \leq 1$. [Hint: M is a direct summand of an induced module.]